MEDICONNECT

Test Cases Execution Report

Frontend | Backend API | AI Analysis Service

Project	MediConnect - patient, doctor, administrator, and AI medical-report platform
Architecture	React/Vite frontend | Express/Mongoose API | FastAPI AI service
Execution date	20 October 2026, 14:27:04 IST
Automated test cases	122 executed: 122 passed, 0 failed, 0 skipped
Additional quality checks	2 npm audits (both returned dependency findings); frontend lint and production build completed
Evidence	Individual Chromium-captured evidence pages sourced from raw runner/audit outputs; raw logs are included

Purpose: provide an evidence-backed account of the executed automated tests, the implemented project areas they cover, and material tests that remain unexecuted.

1. Objectives, scope, and environment

The report is based on source inspection of the React frontend, Express/Mongoose backend, and FastAPI AI service, plus the recorded execution logs listed below. The attached sample PDF was used for its cover/overview, blue-gray striped indexes, detailed case sheets, and per-case evidence placement, not for its unrelated project content.

Implemented modules inspected

Layer	Implemented project scope
Frontend	Public registration/login/OTP; role-protected patient, doctor, and admin workspaces; profiles; community; notifications; records, health, care plans, medication and appointments; emergency access; AI report analyzer/comparison and emergency assistant; video consultation/WebRTC.
Backend API	Authentication and role middleware; users/profile; posts; notifications; patient health/records/upload/timeline/medications/appointments/doctor access/emergency/care plans/analysis; doctor clinical workflows; admin governance/audit/broadcast; Socket.IO signaling.
AI service	FastAPI health, /generate and /analyze-report; report section/metric extraction; validation/completeness; local llama.cpp integration with deterministic fallback for analysis.
Data/external services	Mongoose models and MongoDB; Cloudinary or local upload fallback; local LLM; Gemini emergency assistant; WebRTC signaling.

Frameworks and recorded environment

Area	Frameworks / tools
Frontend	Vitest 5.0.5, React Testing Library 16.3.3, user-event 14.6.8, jsdom 29.1.1, axe-core 4.14.0, Vite 8.2.1; Node 22.19.0 / npm 10.9.3.
Backend	Node built-in node:test + assert, jsonwebtoken; Node 22.19.0 / npm 10.9.3.
AI service	pytest 9.1.1, FastAPI TestClient, FastAPI 0.115.12, HTTPX 0.28.1, Pydantic 2.13.5; Python 3.13.7.
Report / screenshots	Python 3.13, ReportLab 4.4.10, Pillow 12.2.0, Chromium headless. Each case image shows a raw runner/audit excerpt from captured logs.

Exact execution commands

- cd frontend; npm test
- cd backend; npm test
- cd ai-service; python -m pytest -v tests
- cd frontend; npm run lint
- cd frontend; npm run build
- cd backend; npm audit --omit=dev --json
- cd frontend; npm audit --omit=dev --json

Test files executed

- frontend/src/pages/Login.test.jsx
- frontend/src/pages/Register.test.jsx
- frontend/src/components/Input.test.jsx
- frontend/src/components/Button.test.jsx
- frontend/src/services/api.test.js
- frontend/src/nonfunctional/accessibility.performance.test.jsx
- backend/test/auth.test.js
- backend/test/nonfunctional.test.js

- ai-service/tests/test_api.py

2. Execution summary

Functional testing

Category	Executed	Passed	Failed	Not executed
Frontend UI and API client	51	51	0	0
Backend authentication and authorization	51	51	0	0
AI service API and analysis behavior	12	12	0	0
Functional total	114	114	0	0

Non-functional testing and quality checks

Category	Executed	Passed	Failed	Not executed
Accessibility/performance/reliability tests	8	8	0	0
Production dependency audit	2	0	2	0
Frontend lint/build quality checks	2	2	0	0
Non-functional/quality total	12	10	2	0

Overall executed check totals

Category	Executed	Passed	Failed	Not executed
Automated unit/component/API test cases	122	122	0	0
All executed checks, including audits and build/lint	126	124	2	0
Unexecuted scenarios documented separately	0	0	0	19

Functional tests are unit/component or in-process API tests. The React auth context and Axios adapter are mocked. The AI TestClient invokes the real FastAPI app in-process, while the LLM upstream is mocked. No live Express+MongoDB end-to-end workflow was executed.

3. Functional test case index

All rows below correspond to a named test that passed in the recorded final run. Case IDs are report identifiers; the adjacent evidence image reproduces its captured test-runner output line.

Test Case ID	Test Case Name	Module / short description	Status
FE-F-001	renders an input with type text	Shared form input src\components\Input.test.jsx	PASSED
FE-F-002	renders an input with type email	Shared form input src\components\Input.test.jsx	PASSED
FE-F-003	renders an input with type password	Shared form input src\components\Input.test.jsx	PASSED
FE-F-004	renders an input with type number	Shared form input src\components\Input.test.jsx	PASSED
FE-F-005	renders an input with type tel	Shared form input src\components\Input.test.jsx	PASSED
FE-F-006	associates a provided id with its visible label	Shared form input src\components\Input.test.jsx	PASSED
FE-F-007	shows the validation error and error border style	Shared form input src\components\Input.test.jsx	PASSED
FE-F-008	forwards placeholder, value, and change events	Shared form input src\components\Input.test.jsx	PASSED
FE-F-009	builds a development API URL and includes credentials	Axios API client src\services\api.test.js	PASSED
FE-F-010	unwraps the standard successful API response body	Axios API client src\services\api.test.js	PASSED
FE-F-011	normalizes API errors using the server error message	Axios API client src\services\api.test.js	PASSED
FE-F-012	renders the primary variant styles	Shared action button src\components\Button.test.jsx	PASSED
FE-F-013	renders the secondary variant styles	Shared action button src\components\Button.test.jsx	PASSED
FE-F-014	renders the danger variant styles	Shared action button src\components\Button.test.jsx	PASSED
FE-F-015	disables the button when disabled is true	Shared action button src\components\Button.test.jsx	PASSED
FE-F-016	shows a loading indicator and disables the button while loading	Shared action button src\components\Button.test.jsx	PASSED
FE-F-017	appends a caller-provided class name	Shared action button src\components\Button.test.jsx	PASSED
FE-F-018	forwards button type and click handlers	Shared action button src\components\Button.test.jsx	PASSED
FE-F-019	keeps enabled buttons interactive	Shared action button src\components\Button.test.jsx	PASSED
FE-F-020	renders the email and password fields	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-021	requires a name, email, and password	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-022	shows required-field errors without calling the login action	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-023	shows a validation error for malformed email addresses	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-024	rejects passwords shorter than six characters	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-025	logs in and redirects to the dashboard after valid submission	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-026	registers valid details and routes to OTP verification	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-027	offers OTP verification when the backend rejects an unverified account	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-028	rejects malformed email address invalid	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-029	rejects malformed email address invalid	Authentication / Registration src\pages\Register.test.jsx	PASSED

Test Case ID	Test Case Name	Module / short description	Status
FE-F-030	rejects malformed email address missing-at.example.com	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-031	rejects malformed email address user@	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-032	rejects malformed email address missing-at.example.com	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-033	rejects malformed email address @example.com	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-034	rejects malformed email address user@	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-035	rejects malformed email address user@example	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-036	treats whitespace-only email as missing	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-037	requires a password when the email is valid	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-038	clears the email validation message when the email is edited	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-039	rejects malformed email address @example.com	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-040	rejects malformed email address user@example	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-041	rejects a whitespace-only name	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-042	requires a password when name and email are valid	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-043	shows the backend error message when login fails for another reason	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-044	shows the verified confirmation banner from the query string	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-045	clears the name validation message when the name is edited	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-046	returns to the originally requested route after login	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-047	shows the backend registration error	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-048	submits email addresses containing plus tags unchanged	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-049	submits the selected doctor role	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-050	submits the selected admin role	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-051	URL-encodes plus signs in the email when navigating to OTP verification	Authentication / Registration src\pages\Register.test.jsx	PASSED
BE-F-001	rejects registration when required fields are missing	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-002	rejects registration passwords shorter than six characters	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-003	rejects registration with an unsupported role	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-004	requires email and OTP code for verification	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-005	requires credentials for login	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-006	clears the authentication cookie on logout	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-007	allows requests matching an authorized role	Role authorization middleware backend\test\auth.test.js	PASSED
BE-F-008	rejects requests with an unauthorized role	Role authorization middleware backend\test\auth.test.js	PASSED

Test Case ID	Test Case Name	Module / short description	Status
BE-F-009	rejects registration when name omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-010	rejects registration when email omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-011	rejects registration when password omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-012	rejects registration when name and email omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-013	rejects registration when name and password omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-014	rejects registration when email and password omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-015	rejects registration when all required fields omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-016	rejects registration password "a" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-017	rejects registration password "12" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-018	rejects registration password "abc" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-019	rejects registration password "abcd" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-020	rejects registration password "abcde" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-021	rejects registration password " " below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-022	rejects unsupported registration role "superuser"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-023	rejects unsupported registration role "guest"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-024	rejects unsupported registration role "Patient!"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-025	rejects unsupported registration role "doctor-admin"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-026	rejects unsupported registration role "admin "	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-027	rejects unsupported registration role " PATIENT"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-028	rejects unsupported registration role 7	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-029	rejects unsupported registration role {}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-030	requires both verification fields for {}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-031	requires both verification fields for {"email":"patient@example.com"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-032	requires both verification fields for {"otpCode":"123456"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-033	requires both login credentials for {}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-034	requires both login credentials for {"email":"patient@example.com"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-035	requires both login credentials for {"password":"secret123"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-036	allows patient access when permitted by patient	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-037	allows doctor access when permitted by doctor	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-038	allows admin access when permitted by admin	Additional authentication input cases backend\test\auth.test.js	PASSED

Test Case ID	Test Case Name	Module / short description	Status
BE-F-039	allows patient access when permitted by doctor, patient	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-040	allows doctor access when permitted by admin, doctor	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-041	allows admin access when permitted by patient, admin	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-042	denies role access when user {"role":"patient"} is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-043	denies role access when user {"role":"doctor"} is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-044	denies role access when user {"role":"admin"} is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-045	denies role access when user undefined is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-046	rejects protected request with no cookie or authorization header	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-047	rejects protected request with empty authentication cookie	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-048	rejects protected request with unsupported authorization scheme	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-049	rejects protected request with malformed token	JWT authentication checks backend\test\auth.test.js	PASSED
BE-F-050	rejects protected request with token signed by an unknown secret	JWT authentication checks backend\test\auth.test.js	PASSED
BE-F-051	rejects protected request with expired token	JWT authentication checks backend\test\auth.test.js	PASSED
AI-F-001	health endpoint returns service status	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-002	generate rejects missing prompt	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-003	generate rejects whitespace prompt	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-004	generate returns mocked local llm response	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-005	analyze report rejects missing report text	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-006	analyze report rejects whitespace report text	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-007	analyze report extracts and preserves verified metrics	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-008	analysis rejects invalid llm schema	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-009	repair and parse json handles code fences and trailing commas	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-010	repair and parse json rejects empty llm output	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-011	analyze report falls back to deterministic result when llm is offline	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-012	generate returns service unavailable when local llm is offline	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED

4. Functional testing - Frontend

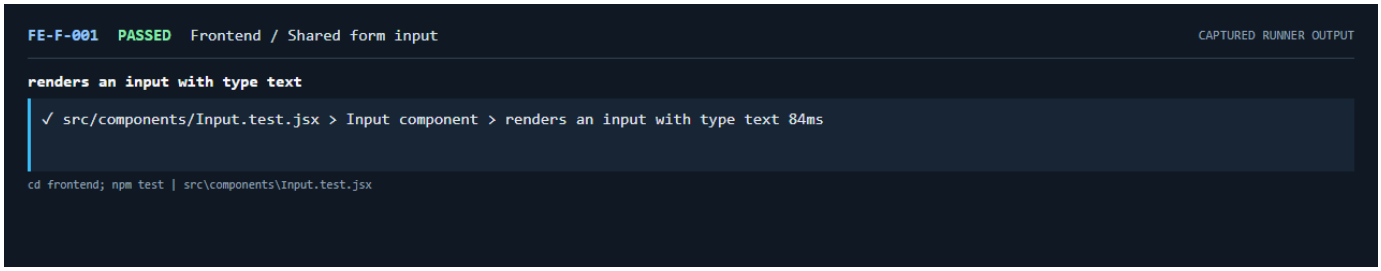
Tests cover login/registration UI validation and navigation, reusable inputs/buttons, and Axios response/error normalization. UI auth context and Axios adapters are stubbed; these are not end-to-end backend calls.

FE-F-001 - renders an input with type text

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: renders an input with type text.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > renders an input with type text 84ms
Execution status	PASSED

Individual execution evidence



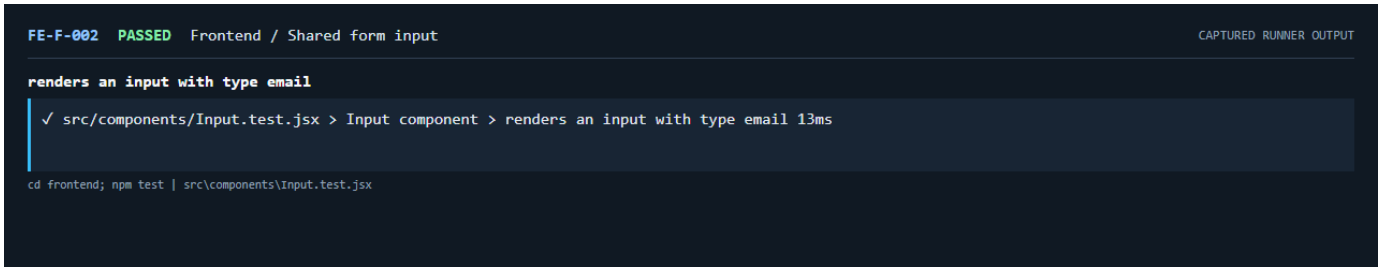
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-001.png.

FE-F-002 - renders an input with type email

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: renders an input with type email.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > renders an input with type email 13ms
Execution status	PASSED

Individual execution evidence



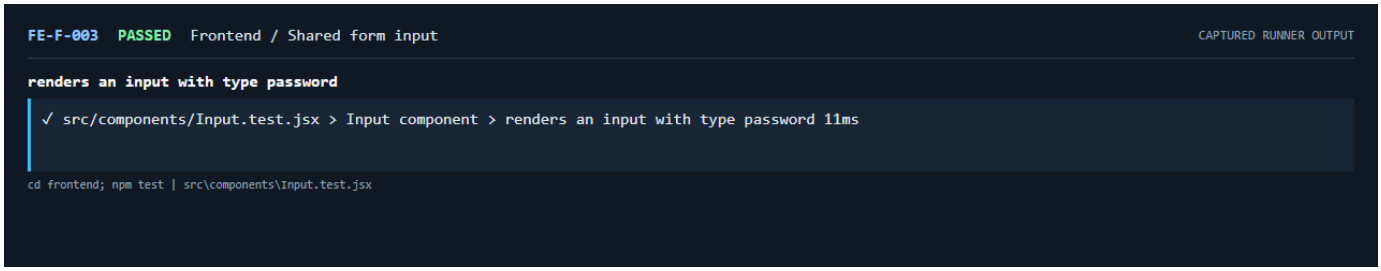
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-002.png.

FE-F-003 - renders an input with type password

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: renders an input with type password.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > renders an input with type password 11ms
Execution status	PASSED

Individual execution evidence



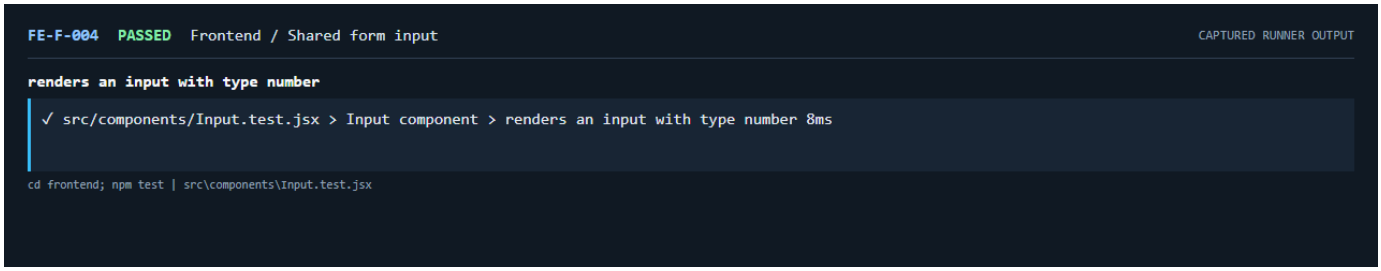
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-003.png.

FE-F-004 - renders an input with type number

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: renders an input with type number.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > renders an input with type number 8ms
Execution status	PASSED

Individual execution evidence



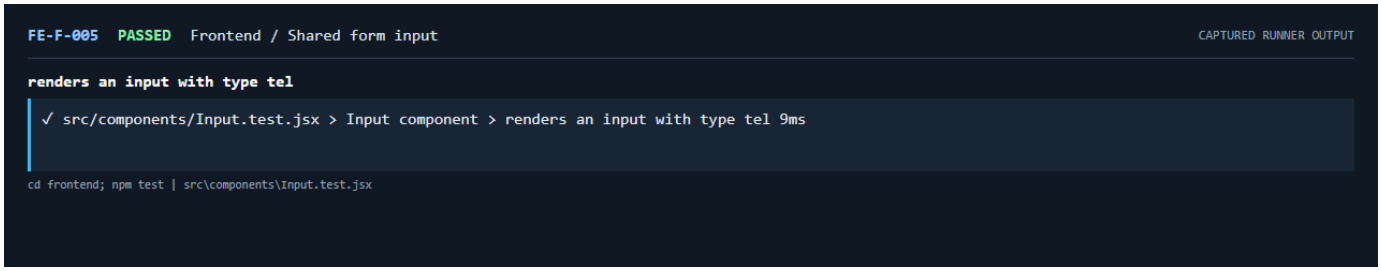
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-004.png.

FE-F-005 - renders an input with type tel

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: renders an input with type tel.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > renders an input with type tel 9ms
Execution status	PASSED

Individual execution evidence



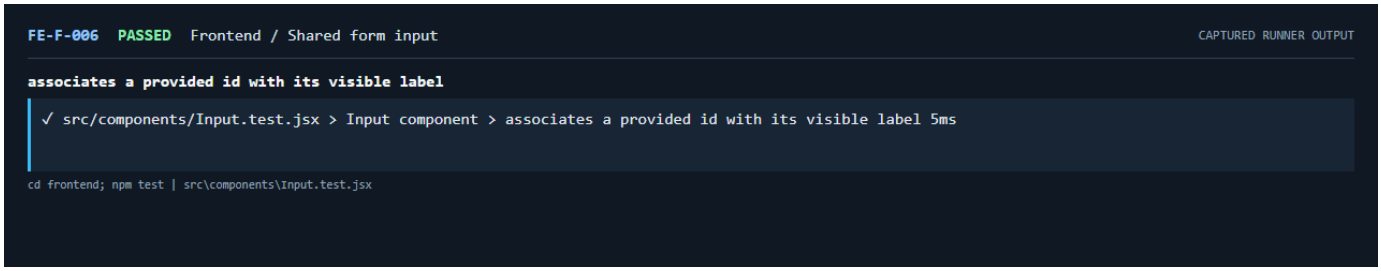
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-005.png.

FE-F-006 - associates a provided id with its visible label

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: associates a provided id with its visible label.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > associates a provided id with its visible label 5ms
Execution status	PASSED

Individual execution evidence



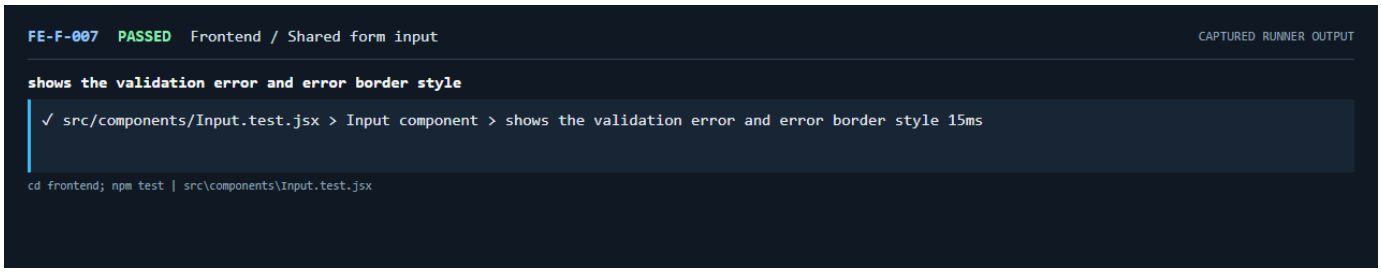
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-006.png.

FE-F-007 - shows the validation error and error border style

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: shows the validation error and error border style.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > shows the validation error and error border style 15ms
Execution status	PASSED

Individual execution evidence



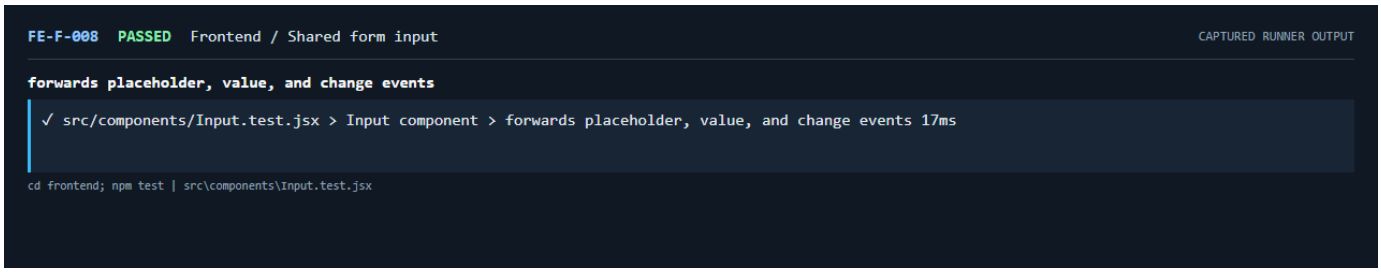
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-007.png.

FE-F-008 - forwards placeholder, value, and change events

PASSED | Frontend | Functional

Feature / module	Shared form input
Requirement / purpose	Verify the implemented behavior: forwards placeholder, value, and change events.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic label, input type, placeholder, value, and event handler.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Input.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Input.test.jsx > Input component > forwards placeholder, value, and change events 17ms
Execution status	PASSED

Individual execution evidence



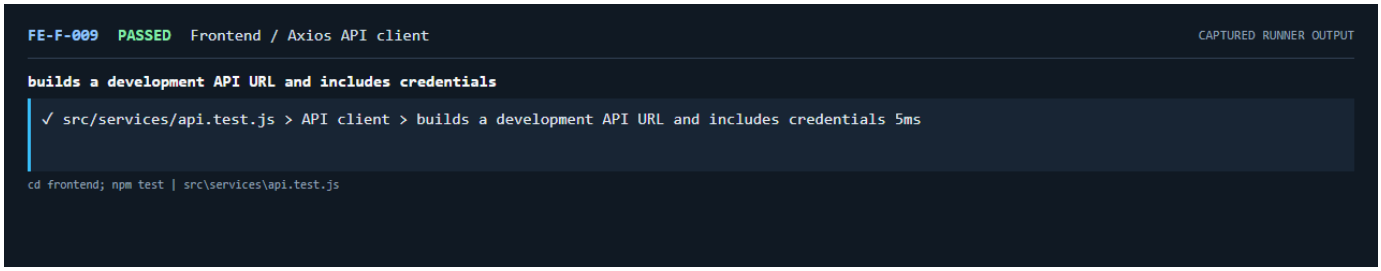
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-008.png.

FE-F-009 - builds a development API URL and includes credentials

PASSED | Frontend | Functional

Feature / module	Axios API client
Requirement / purpose	Verify the implemented behavior: builds a development API URL and includes credentials.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic Axios adapter response/error; no network request is sent.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\services\api.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/services/api.test.js > API client > builds a development API URL and includes credentials 5ms
Execution status	PASSED

Individual execution evidence



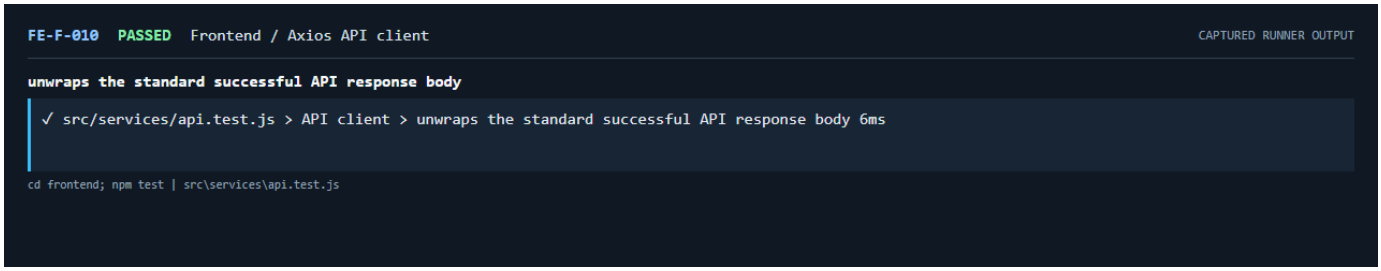
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-009.png.

FE-F-010 - unwraps the standard successful API response body

PASSED | Frontend | Functional

Feature / module	Axios API client
Requirement / purpose	Verify the implemented behavior: unwraps the standard successful API response body.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic Axios adapter response/error; no network request is sent.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\services\api.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/services/api.test.js > API client > unwraps the standard successful API response body 6ms
Execution status	PASSED

Individual execution evidence



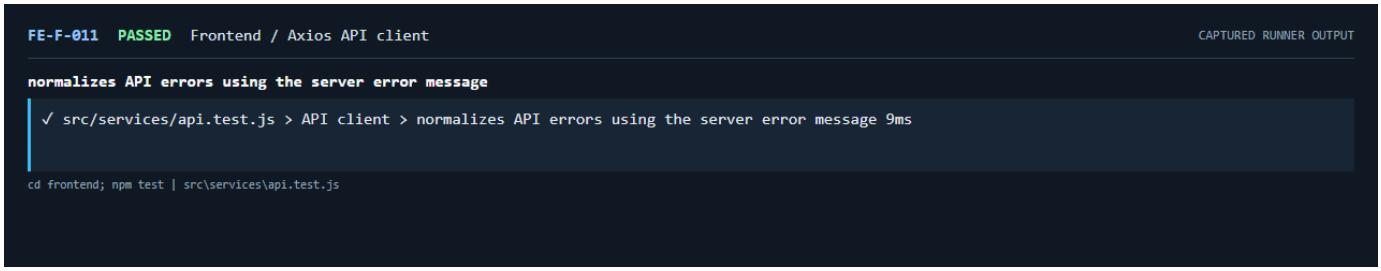
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-010.png.

FE-F-011 - normalizes API errors using the server error message

PASSED | Frontend | Functional

Feature / module	Axios API client
Requirement / purpose	Verify the implemented behavior: normalizes API errors using the server error message.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic Axios adapter response/error; no network request is sent.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\services\api.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/services/api.test.js > API client > normalizes API errors using the server error message 9ms
Execution status	PASSED

Individual execution evidence



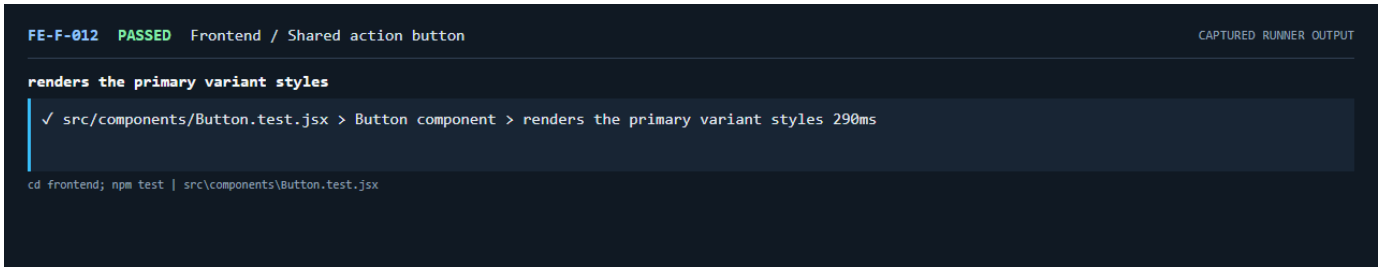
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-011.png.

FE-F-012 - renders the primary variant styles

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: renders the primary variant styles.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > renders the primary variant styles 290ms
Execution status	PASSED

Individual execution evidence



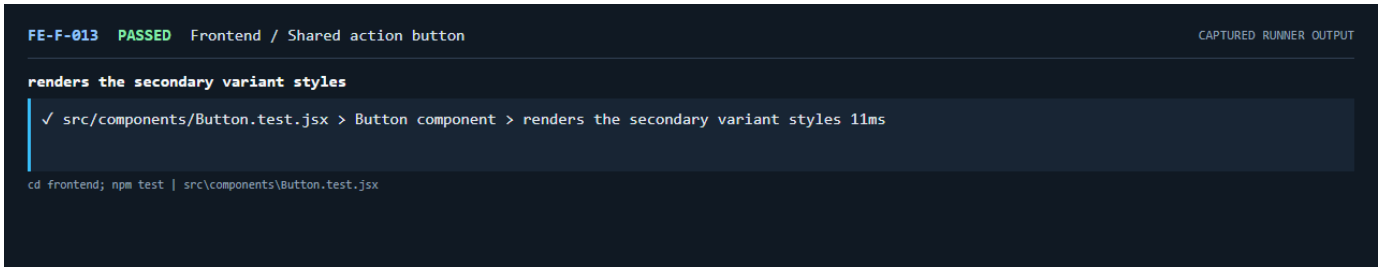
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-012.png.

FE-F-013 - renders the secondary variant styles

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: renders the secondary variant styles.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > renders the secondary variant styles 11ms
Execution status	PASSED

Individual execution evidence



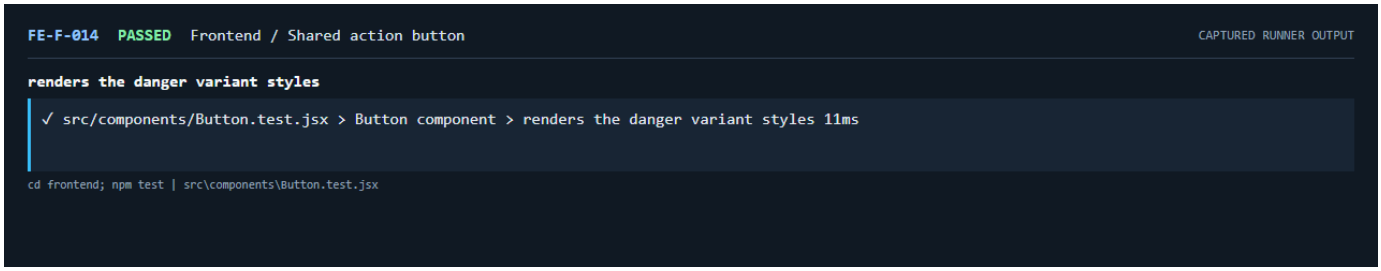
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-013.png.

FE-F-014 - renders the danger variant styles

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: renders the danger variant styles.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > renders the danger variant styles 11ms
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-014.png.

FE-F-015 - disables the button when disabled is true

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: disables the button when disabled is true.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > disables the button when disabled is true 8ms
Execution status	PASSED

Individual execution evidence



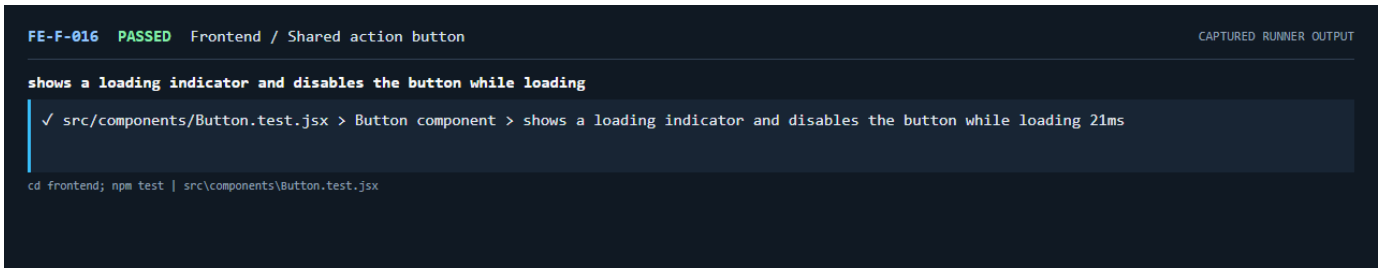
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-015.png.

FE-F-016 - shows a loading indicator and disables the button while loading

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: shows a loading indicator and disables the button while loading.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > shows a loading indicator and disables the button while loading 21ms
Execution status	PASSED

Individual execution evidence



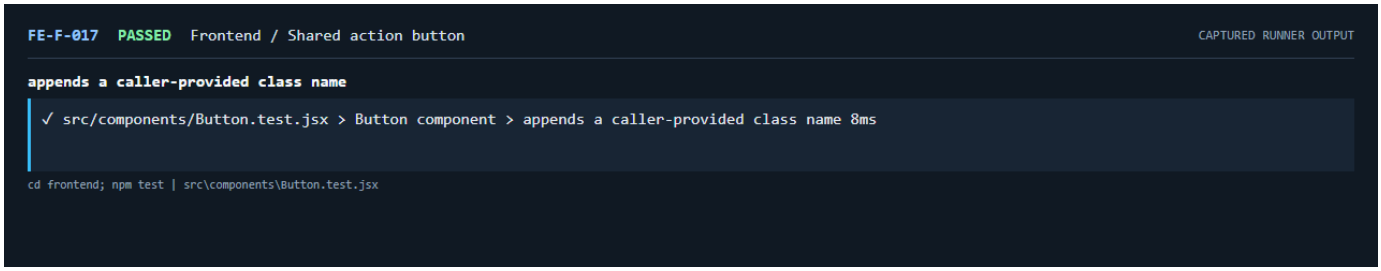
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-016.png.

FE-F-017 - appends a caller-provided class name

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: appends a caller-provided class name.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > appends a caller-provided class name 8ms
Execution status	PASSED

Individual execution evidence



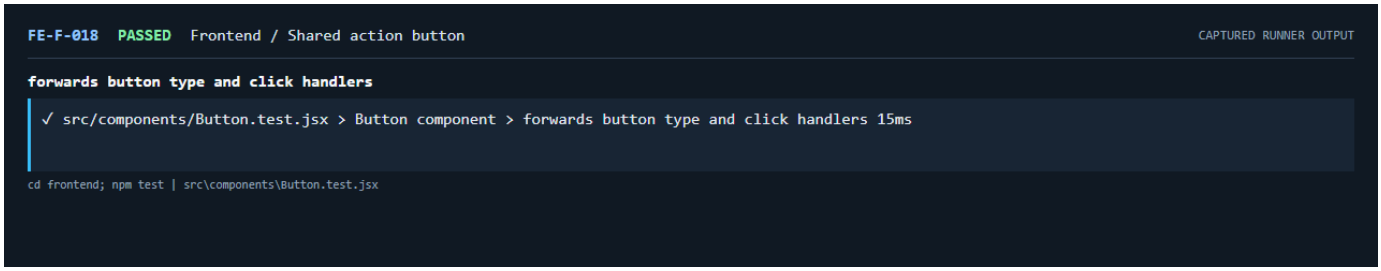
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-017.png.

FE-F-018 - forwards button type and click handlers

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: forwards button type and click handlers.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > forwards button type and click handlers 15ms
Execution status	PASSED

Individual execution evidence



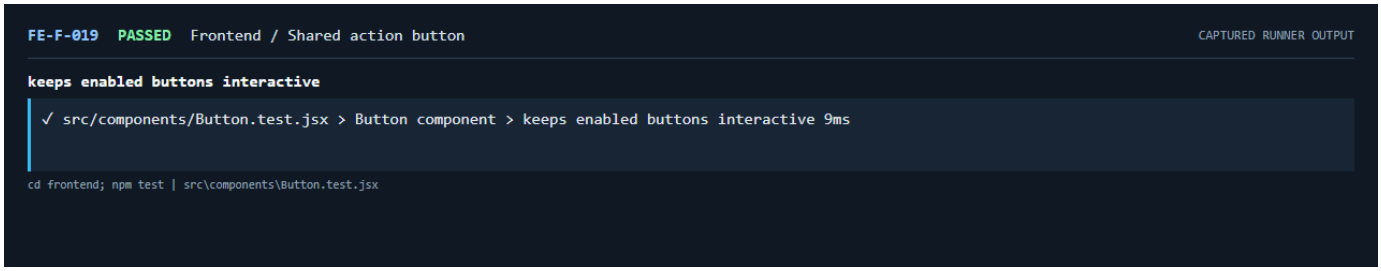
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-018.png.

FE-F-019 - keeps enabled buttons interactive

PASSED | Frontend | Functional

Feature / module	Shared action button
Requirement / purpose	Verify the implemented behavior: keeps enabled buttons interactive.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic button children, variant, disabled/loading props, and click callback.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\components\Button.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/components/Button.test.jsx > Button component > keeps enabled buttons interactive 9ms
Execution status	PASSED

Individual execution evidence



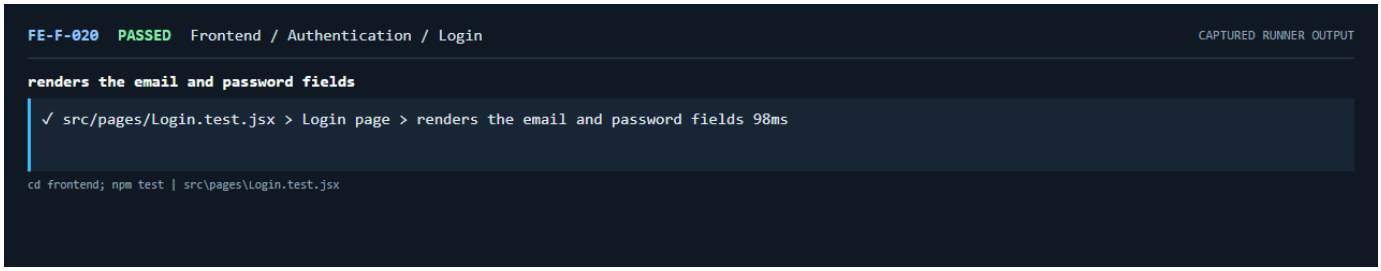
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-019.png.

FE-F-020 - renders the email and password fields

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: renders the email and password fields.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > renders the email and password fields 98ms
Execution status	PASSED

Individual execution evidence



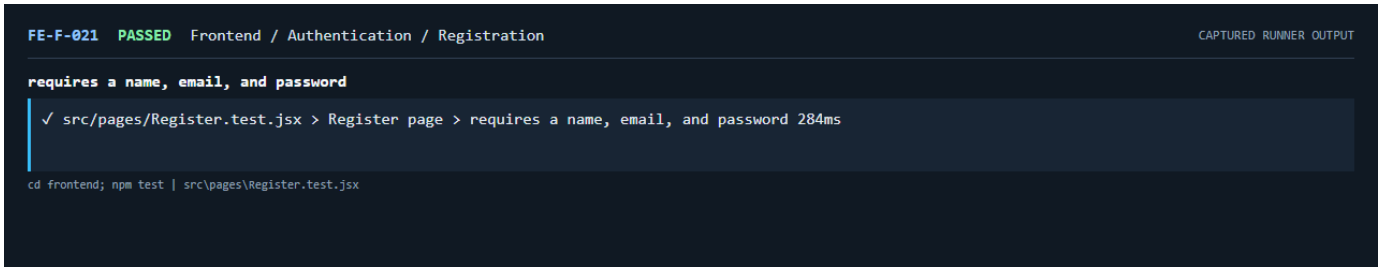
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-020.png.

FE-F-021 - requires a name, email, and password

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: requires a name, email, and password.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > requires a name, email, and password 284ms
Execution status	PASSED

Individual execution evidence



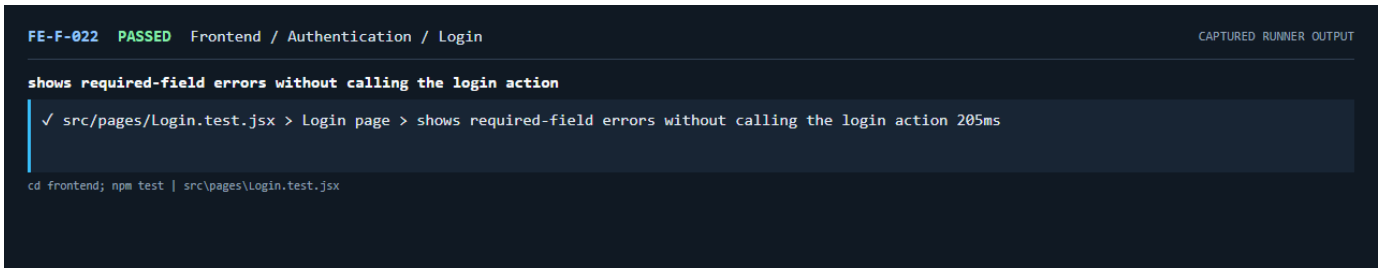
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-021.png.

FE-F-022 - shows required-field errors without calling the login action

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: shows required-field errors without calling the login action.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > shows required-field errors without calling the login action 205ms
Execution status	PASSED

Individual execution evidence



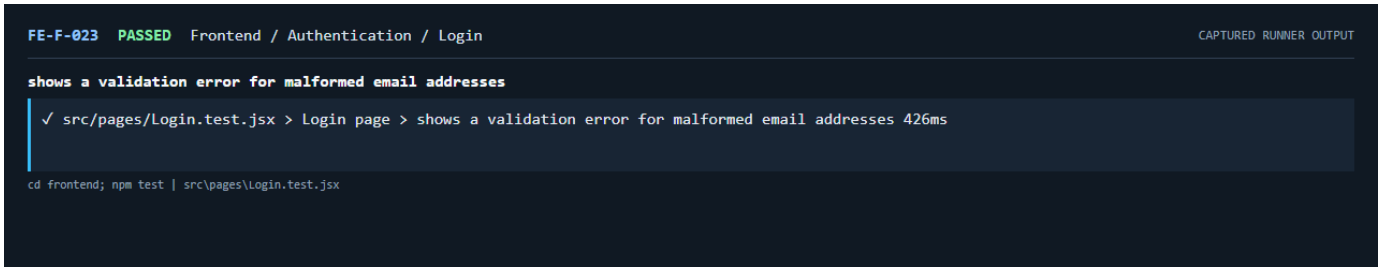
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-022.png.

FE-F-023 - shows a validation error for malformed email addresses

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: shows a validation error for malformed email addresses.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'addresses'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > shows a validation error for malformed email addresses 426ms
Execution status	PASSED

Individual execution evidence



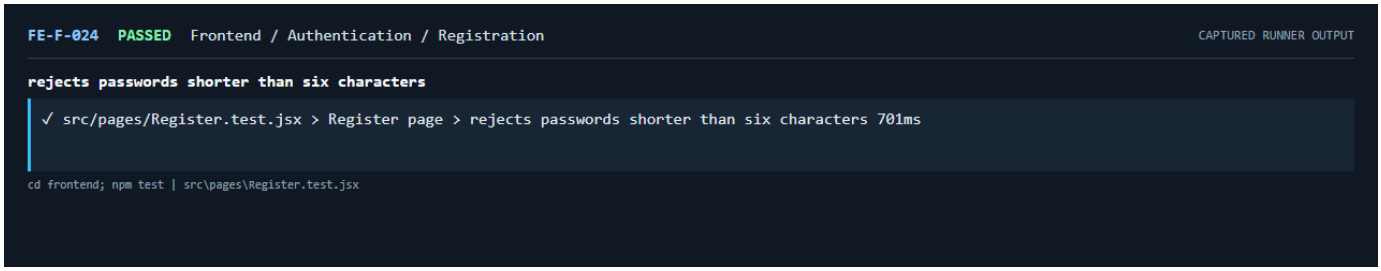
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-023.png.

FE-F-024 - rejects passwords shorter than six characters

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects passwords shorter than six characters.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects passwords shorter than six characters 701ms
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-024.png.

FE-F-025 - logs in and redirects to the dashboard after valid submission

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: logs in and redirects to the dashboard after valid submission.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > logs in and redirects to the dashboard after valid submission 590ms
Execution status	PASSED

Individual execution evidence



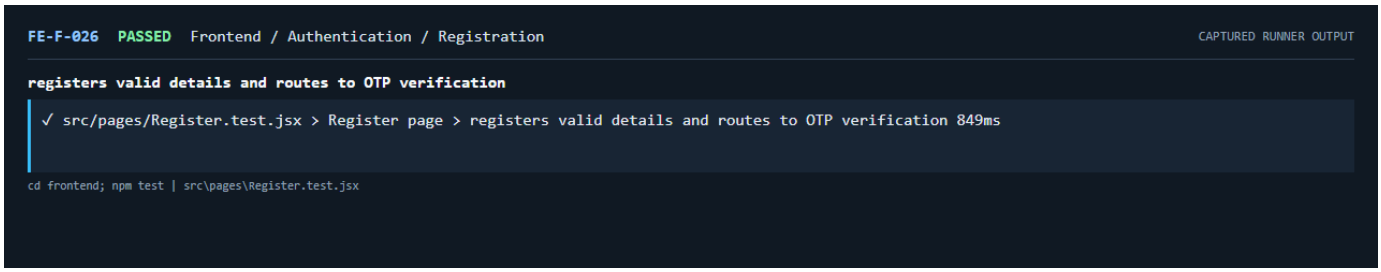
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-025.png.

FE-F-026 - registers valid details and routes to OTP verification

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: registers valid details and routes to OTP verification.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Registration action receives the specified synthetic values and navigation proceeds to encoded OTP verification.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > registers valid details and routes to OTP verification 849ms
Execution status	PASSED

Individual execution evidence



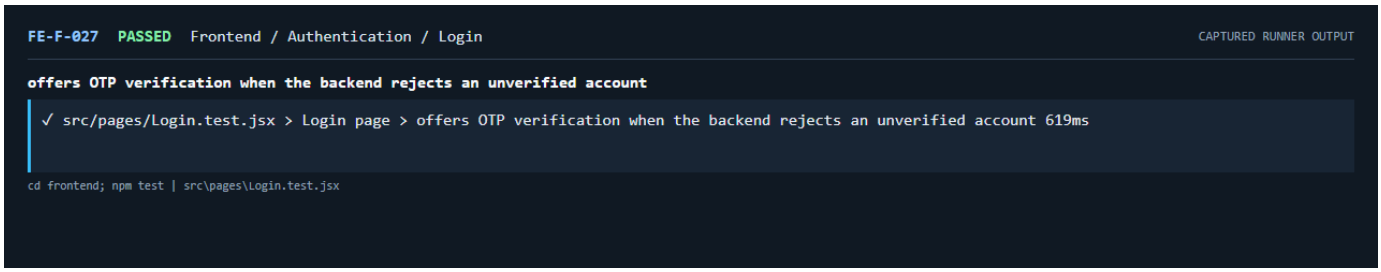
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-026.png.

FE-F-027 - offers OTP verification when the backend rejects an unverified account

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: offers OTP verification when the backend rejects an unverified account.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > offers OTP verification when the backend rejects an unverified account 619ms
Execution status	PASSED

Individual execution evidence



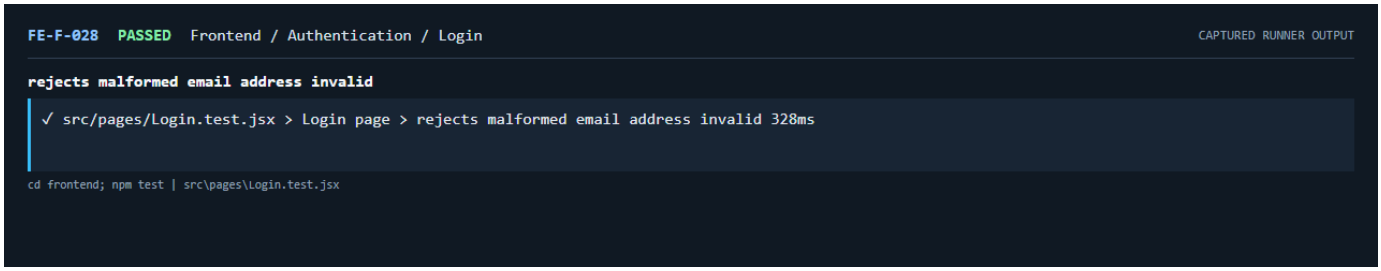
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-027.png.

FE-F-028 - rejects malformed email address invalid

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: rejects malformed email address invalid.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'invalid'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > rejects malformed email address invalid 328ms
Execution status	PASSED

Individual execution evidence



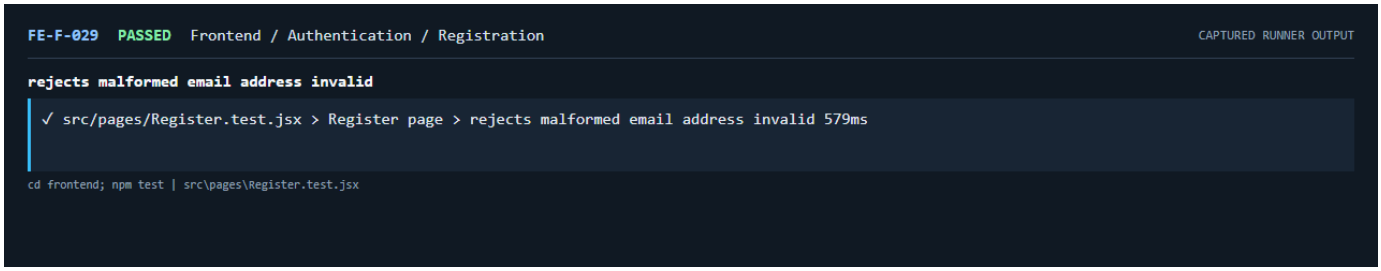
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-028.png.

FE-F-029 - rejects malformed email address invalid

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects malformed email address invalid.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'invalid'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects malformed email address invalid 579ms
Execution status	PASSED

Individual execution evidence



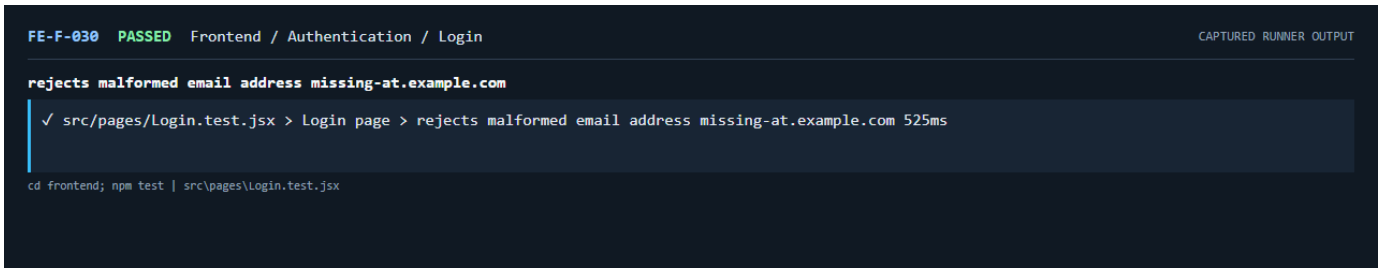
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-029.png.

FE-F-030 - rejects malformed email address missing-at.example.com

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: rejects malformed email address missing-at.example.com.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'missing-at.example.com'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > rejects malformed email address missing-at.example.com 525ms
Execution status	PASSED

Individual execution evidence



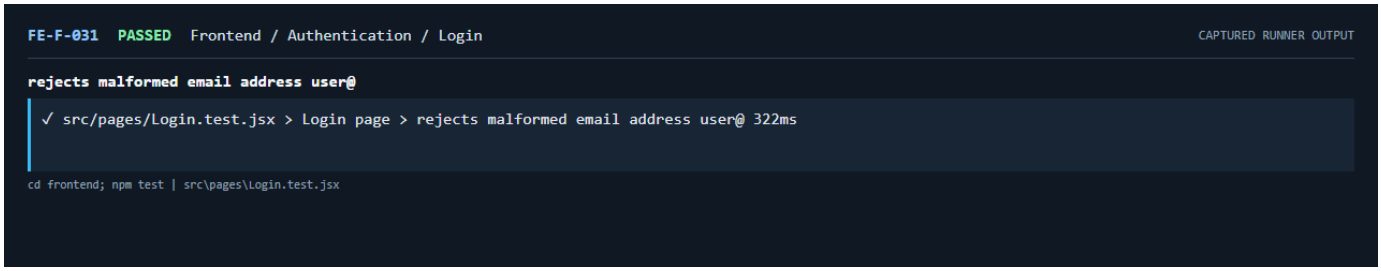
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-030.png.

FE-F-031 - rejects malformed email address user@

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: rejects malformed email address user@.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'user@'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > rejects malformed email address user@ 322ms
Execution status	PASSED

Individual execution evidence



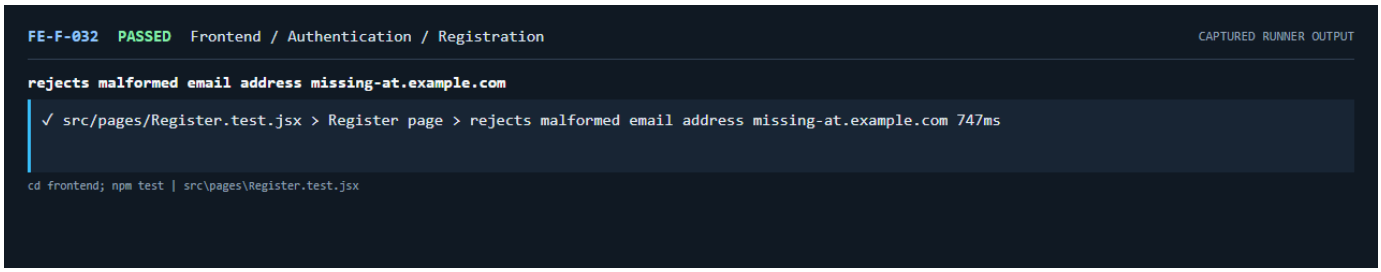
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-031.png.

FE-F-032 - rejects malformed email address missing-at.example.com

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects malformed email address missing-at.example.com.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'missing-at.example.com'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects malformed email address missing-at.example.com 747ms
Execution status	PASSED

Individual execution evidence



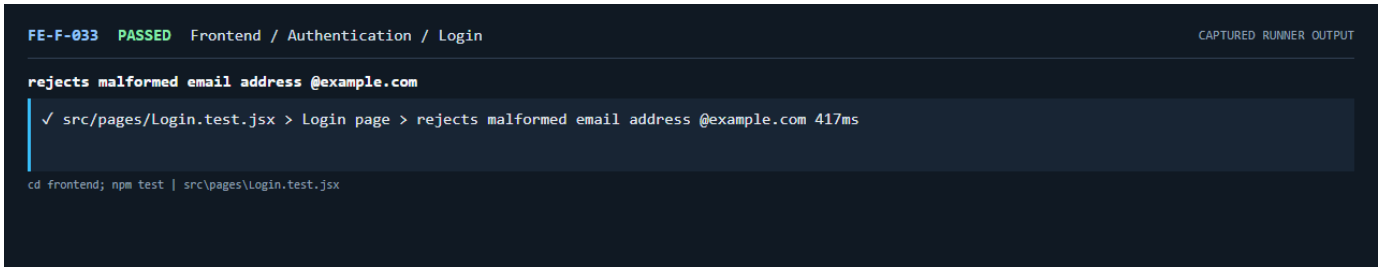
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-032.png.

FE-F-033 - rejects malformed email address @example.com

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: rejects malformed email address @example.com.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: '@example.com'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > rejects malformed email address @example.com 417ms
Execution status	PASSED

Individual execution evidence



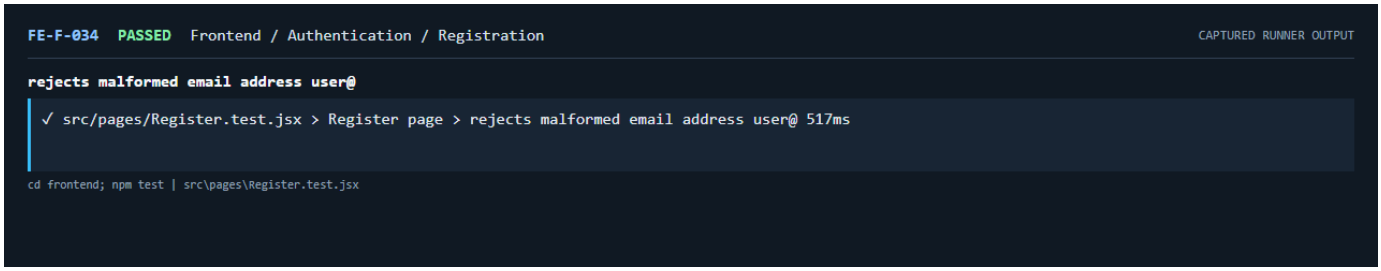
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-033.png.

FE-F-034 - rejects malformed email address user@

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects malformed email address user@.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'user@'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects malformed email address user@ 517ms
Execution status	PASSED

Individual execution evidence



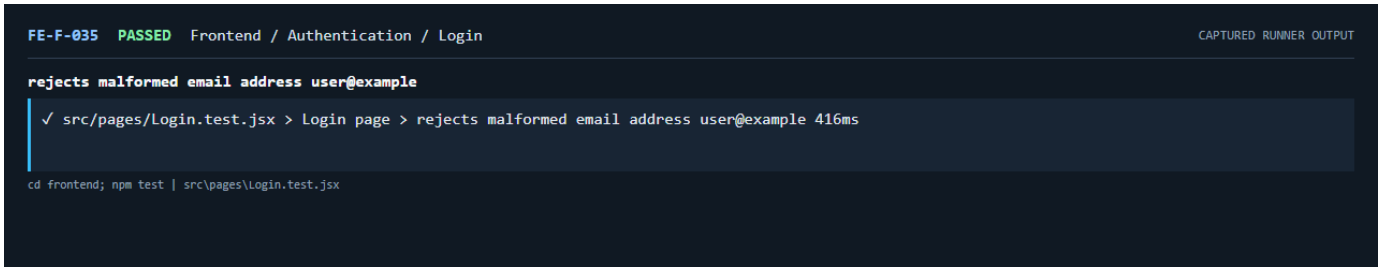
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-034.png.

FE-F-035 - rejects malformed email address user@example

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: rejects malformed email address user@example.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'user@example'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > rejects malformed email address user@example 416ms
Execution status	PASSED

Individual execution evidence



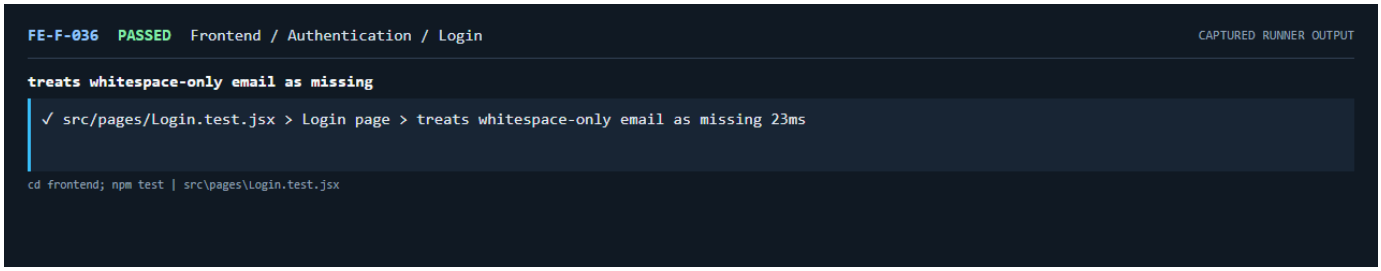
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-035.png.

FE-F-036 - treats whitespace-only email as missing

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: treats whitespace-only email as missing.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email: spaces only; password: secret123.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > treats whitespace-only email as missing 23ms
Execution status	PASSED

Individual execution evidence



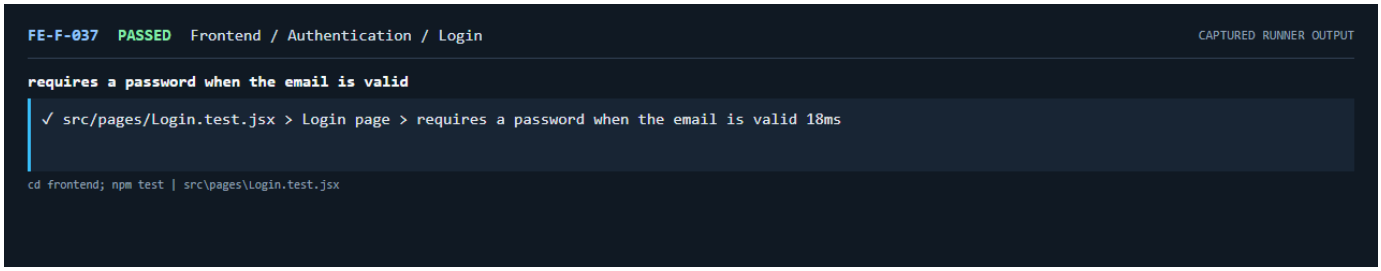
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-036.png.

FE-F-037 - requires a password when the email is valid

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: requires a password when the email is valid.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > requires a password when the email is valid 18ms
Execution status	PASSED

Individual execution evidence



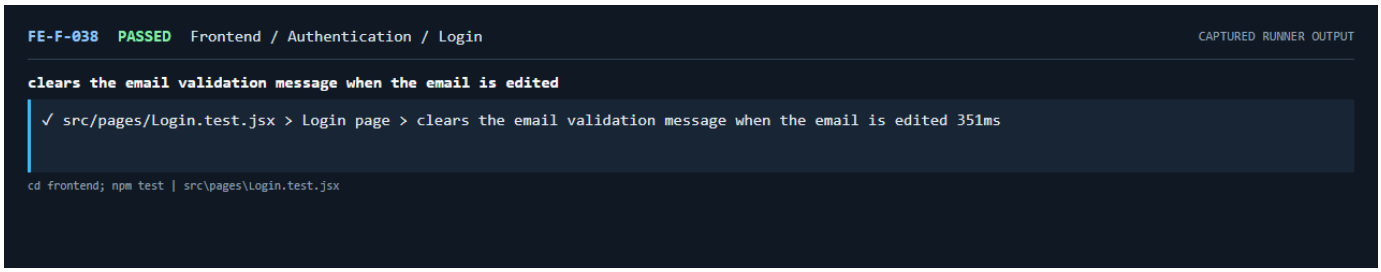
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-037.png.

FE-F-038 - clears the email validation message when the email is edited

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: clears the email validation message when the email is edited.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > clears the email validation message when the email is edited 351ms
Execution status	PASSED

Individual execution evidence



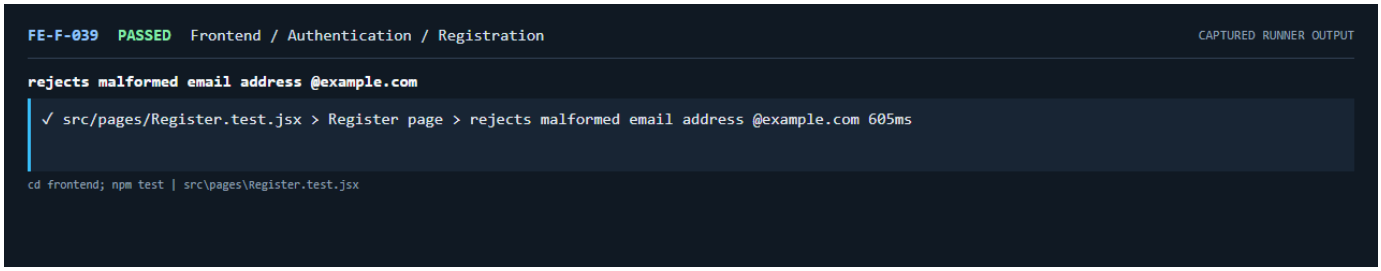
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-038.png.

FE-F-039 - rejects malformed email address @example.com

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects malformed email address @example.com.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: '@example.com'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects malformed email address @example.com 605ms
Execution status	PASSED

Individual execution evidence



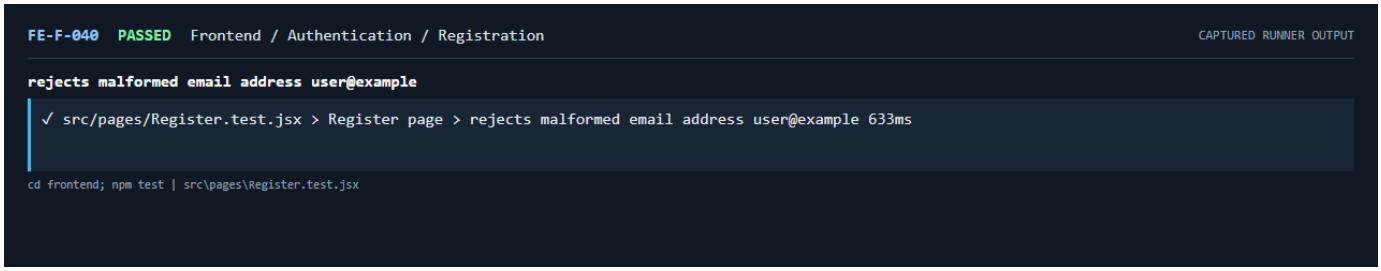
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-039.png.

FE-F-040 - rejects malformed email address user@example

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects malformed email address user@example.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email value: 'user@example'; other fields use valid test values.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Form displays the email-format error and does not call the auth action.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects malformed email address user@example 633ms
Execution status	PASSED

Individual execution evidence



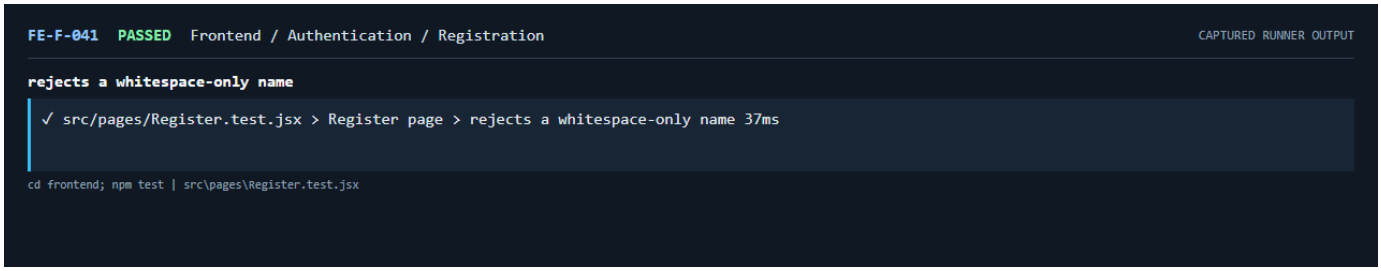
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-040.png.

FE-F-041 - rejects a whitespace-only name

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: rejects a whitespace-only name.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Name: spaces only; valid email and a password longer than five characters.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > rejects a whitespace-only name 37ms
Execution status	PASSED

Individual execution evidence



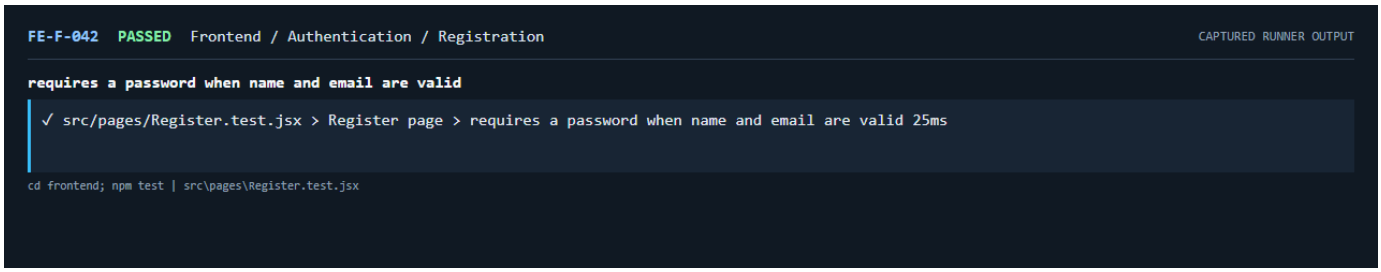
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-041.png.

FE-F-042 - requires a password when name and email are valid

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: requires a password when name and email are valid.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > requires a password when name and email are valid 25ms
Execution status	PASSED

Individual execution evidence



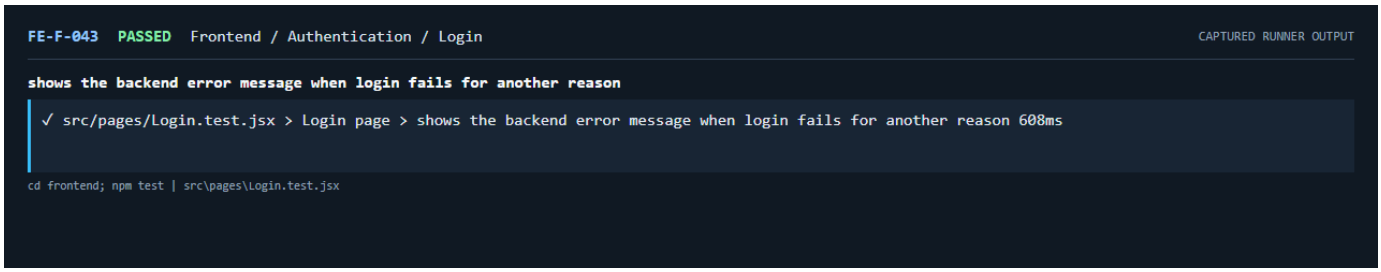
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-042.png.

FE-F-043 - shows the backend error message when login fails for another reason

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: shows the backend error message when login fails for another reason.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Valid-shaped email/password; mocked auth action rejects with an error.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > shows the backend error message when login fails for another reason 608ms
Execution status	PASSED

Individual execution evidence



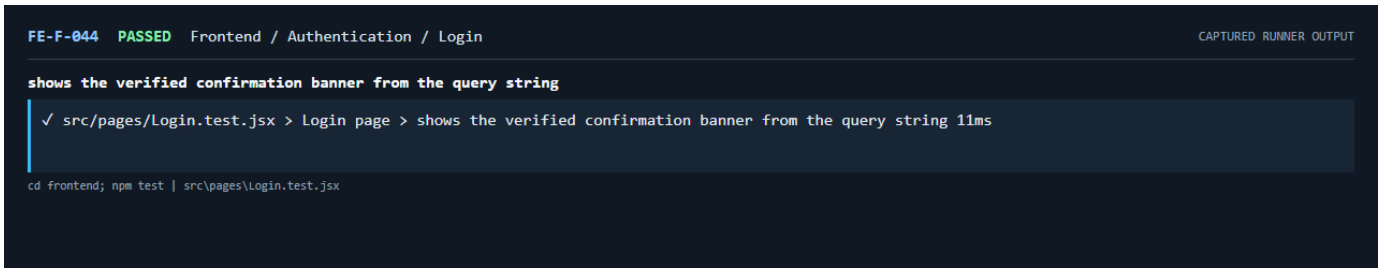
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-043.png.

FE-F-044 - shows the verified confirmation banner from the query string

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: shows the verified confirmation banner from the query string.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > shows the verified confirmation banner from the query string 11ms
Execution status	PASSED

Individual execution evidence



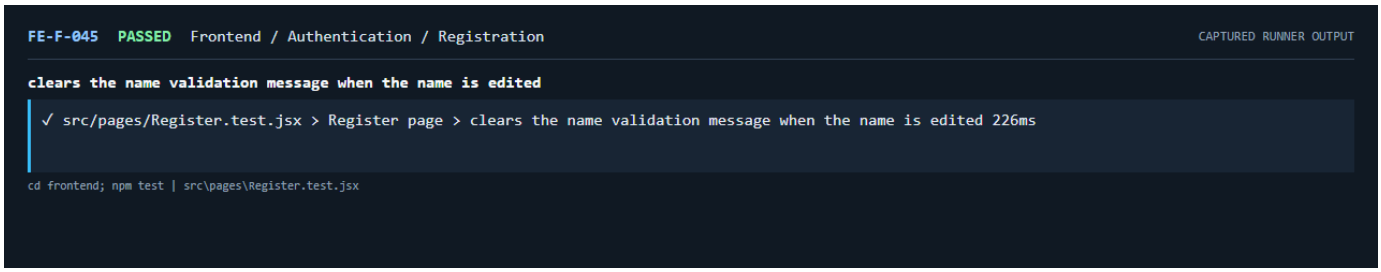
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-044.png.

FE-F-045 - clears the name validation message when the name is edited

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: clears the name validation message when the name is edited.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > clears the name validation message when the name is edited 226ms
Execution status	PASSED

Individual execution evidence



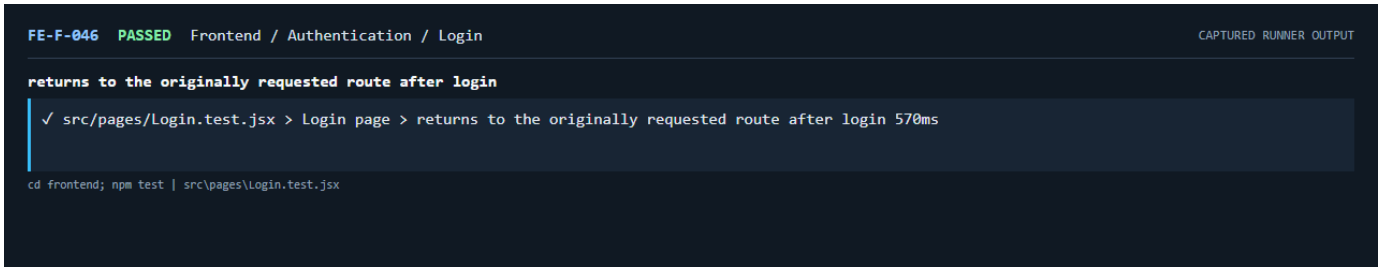
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-045.png.

FE-F-046 - returns to the originally requested route after login

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: returns to the originally requested route after login.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic form values in the test source; auth context is mocked.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > returns to the originally requested route after login 570ms
Execution status	PASSED

Individual execution evidence



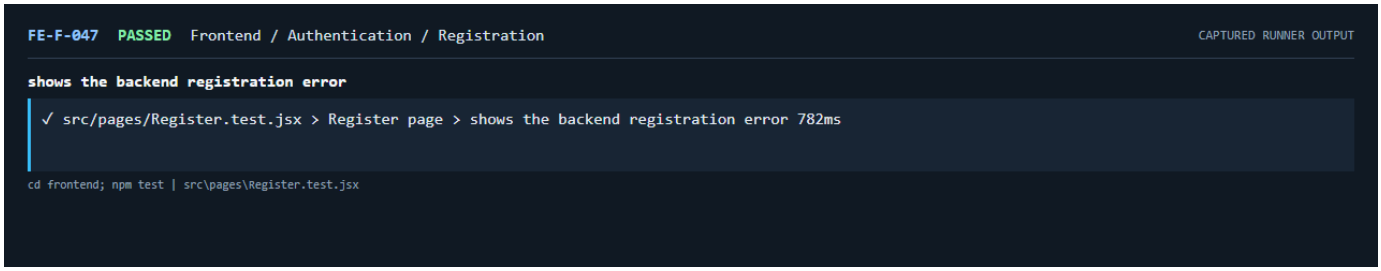
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-046.png.

FE-F-047 - shows the backend registration error

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: shows the backend registration error.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > shows the backend registration error 782ms
Execution status	PASSED

Individual execution evidence



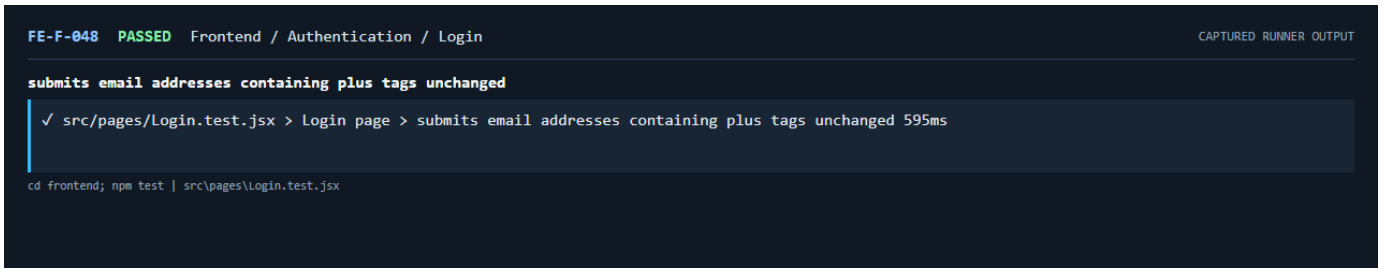
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-047.png.

FE-F-048 - submits email addresses containing plus tags unchanged

PASSED | Frontend | Functional

Feature / module	Authentication / Login
Requirement / purpose	Verify the implemented behavior: submits email addresses containing plus tags unchanged.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email: patient+care@example.com; password: secret123.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Login.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Login.test.jsx > Login page > submits email addresses containing plus tags unchanged 595ms
Execution status	PASSED

Individual execution evidence



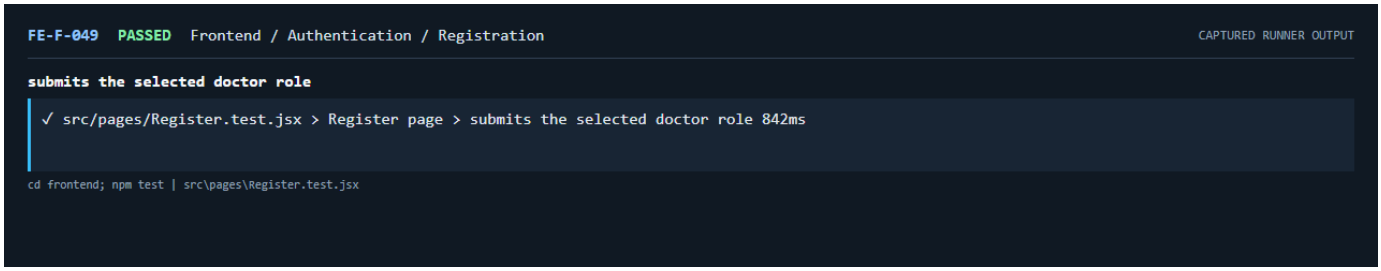
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-048.png.

FE-F-049 - submits the selected doctor role

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: submits the selected doctor role.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic doctor name/email, password secret123, selected role doctor.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > submits the selected doctor role 842ms
Execution status	PASSED

Individual execution evidence



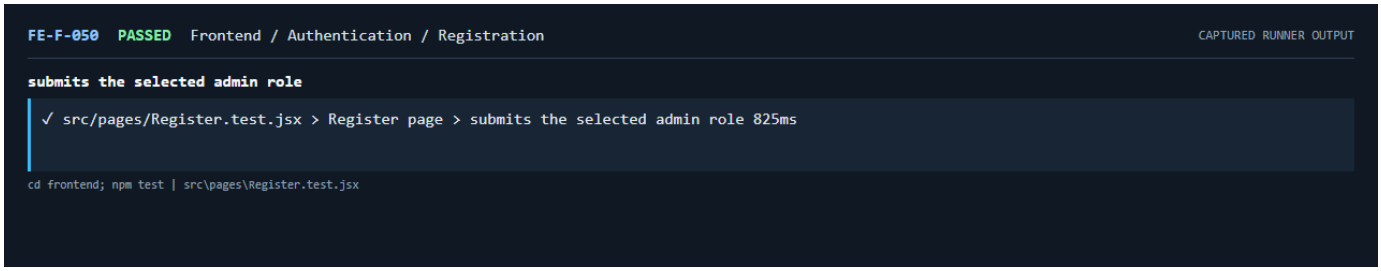
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-049.png.

FE-F-050 - submits the selected admin role

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: submits the selected admin role.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic admin name/email, password secret123, selected role admin.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > submits the selected admin role 825ms
Execution status	PASSED

Individual execution evidence



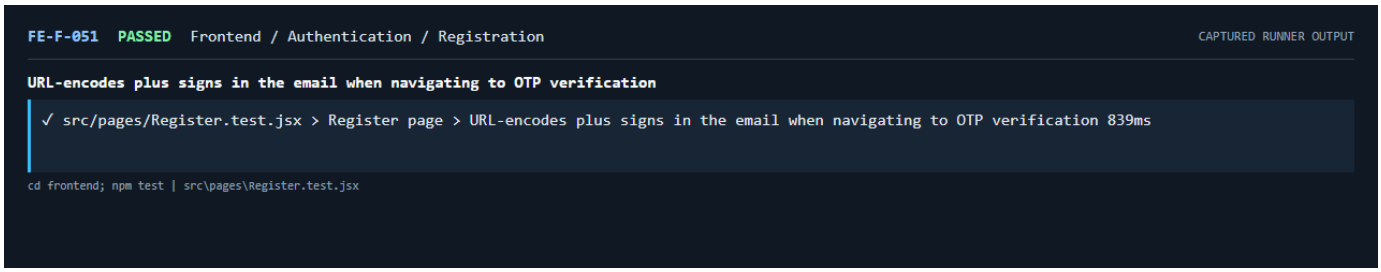
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-050.png.

FE-F-051 - URL-encodes plus signs in the email when navigating to OTP verification

PASSED | Frontend | Functional

Feature / module	Authentication / Registration
Requirement / purpose	Verify the implemented behavior: URL-encodes plus signs in the email when navigating to OTP verification.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Email: patient+care@example.com; password: secret123.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\pages\Register.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✓ src/pages/Register.test.jsx > Register page > URL-encodes plus signs in the email when navigating to OTP verification 839ms
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-F-051.png.

4. Functional testing - Backend

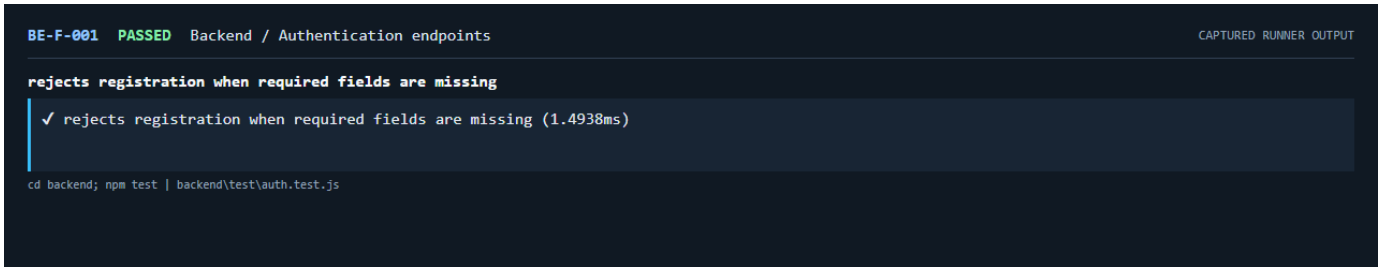
Tests call authentication handlers and JWT/role middleware directly with mock Express response objects. They cover validation, logout cookie behavior, allowed/denied roles, and malformed/expired/invalid-signature JWT rejection. Database-backed paths are not reached.

BE-F-001 - rejects registration when required fields are missing

PASSED | Backend | Functional

Feature / module	Authentication endpoints
Requirement / purpose	Verify the implemented behavior: rejects registration when required fields are missing.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when required fields are missing (1.4938ms)
Execution status	PASSED

Individual execution evidence



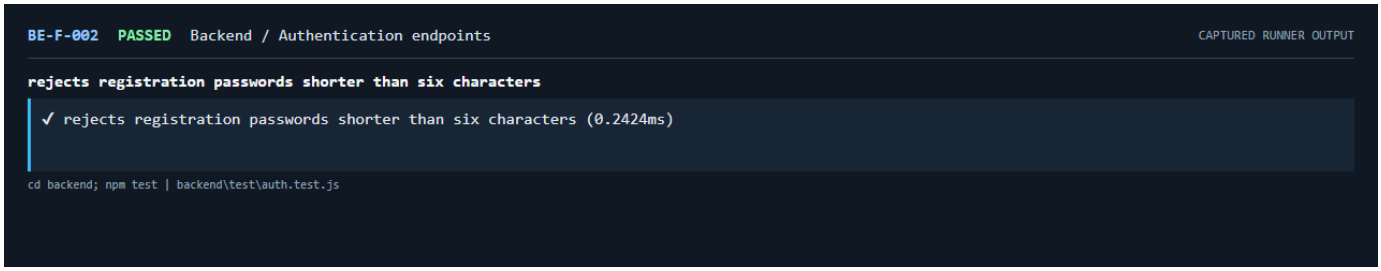
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-001.png.

BE-F-002 - rejects registration passwords shorter than six characters

PASSED | Backend | Functional

Feature / module	Authentication endpoints
Requirement / purpose	Verify the implemented behavior: rejects registration passwords shorter than six characters.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects registration passwords shorter than six characters (0.2424ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-002.png.

BE-F-003 - rejects registration with an unsupported role

PASSED | Backend | Functional

Feature / module	Authentication endpoints
Requirement / purpose	Verify the implemented behavior: rejects registration with an unsupported role.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects registration with an unsupported role (0.205ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-003.png.

BE-F-004 - requires email and OTP code for verification

PASSED | Backend | Functional

Feature / module	Authentication endpoints
Requirement / purpose	Verify the implemented behavior: requires email and OTP code for verification.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires email and OTP code for verification (0.3927ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-004.png.

BE-F-005 - requires credentials for login

PASSED | Backend | Functional

Feature / module	Authentication endpoints
Requirement / purpose	Verify the implemented behavior: requires credentials for login.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires credentials for login (0.2719ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-005.png.

BE-F-006 - clears the authentication cookie on logout

PASSED | Backend | Functional

Feature / module	Authentication endpoints
Requirement / purpose	Verify the implemented behavior: clears the authentication cookie on logout.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Mock response object; no browser cookie jar or external service.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Response confirms logout and clears the HTTP-only JWT cookie.
Actual result	Captured runner result: ✔ clears the authentication cookie on logout (0.5981ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-006.png.

BE-F-007 - allows requests matching an authorized role

PASSED | Backend | Functional

Feature / module	Role authorization middleware
Requirement / purpose	Verify the implemented behavior: allows requests matching an authorized role.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ allows requests matching an authorized role (0.443ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-007.png.

BE-F-008 - rejects requests with an unauthorized role

PASSED | Backend | Functional

Feature / module	Role authorization middleware
Requirement / purpose	Verify the implemented behavior: rejects requests with an unauthorized role.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Request is rejected with HTTP 403; next handler is not called.
Actual result	Captured runner result: ✔ rejects requests with an unauthorized role (0.5869ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-008.png.

BE-F-009 - rejects registration when name omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when name omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: name; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when name omitted (1.1982ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-009.png.

BE-F-010 - rejects registration when email omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when email omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: email; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when email omitted (0.9694ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-010.png.

BE-F-011 - rejects registration when password omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when password omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: password; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when password omitted (0.2685ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-011.png.

BE-F-012 - rejects registration when name and email omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when name and email omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: name and email; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when name and email omitted (0.2186ms)
Execution status	PASSED

Individual execution evidence



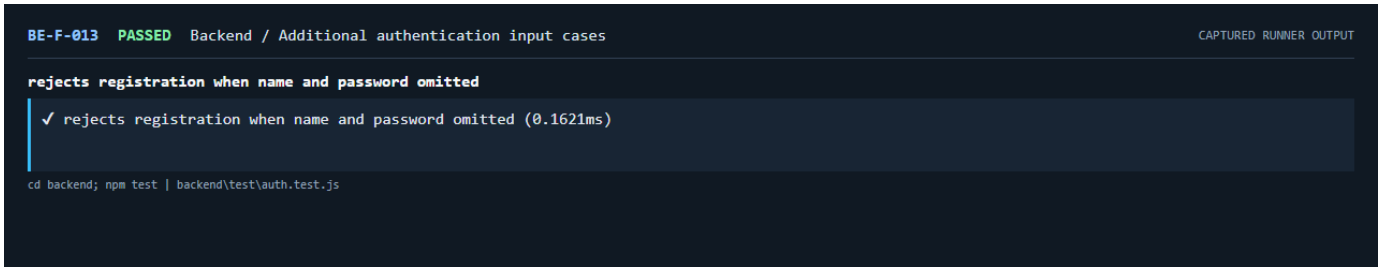
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-012.png.

BE-F-013 - rejects registration when name and password omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when name and password omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: name and password; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when name and password omitted (0.1621ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-013.png.

BE-F-014 - rejects registration when email and password omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when email and password omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: email and password; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when email and password omitted (0.1698ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-014.png.

BE-F-015 - rejects registration when all required fields omitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration when all required fields omitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration body omits: all required fields; remaining required fields use valid synthetic values.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects registration when all required fields omitted (0.392ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-015.png.

BE-F-016 - rejects registration password "a" below minimum length

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration password "a" below minimum length.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration password is the boundary value embedded in the case name: "a" below minimum length.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Password boundary is rejected with validation feedback and no auth action.
Actual result	Captured runner result: ✔ rejects registration password "a" below minimum length (0.2827ms)
Execution status	PASSED

Individual execution evidence



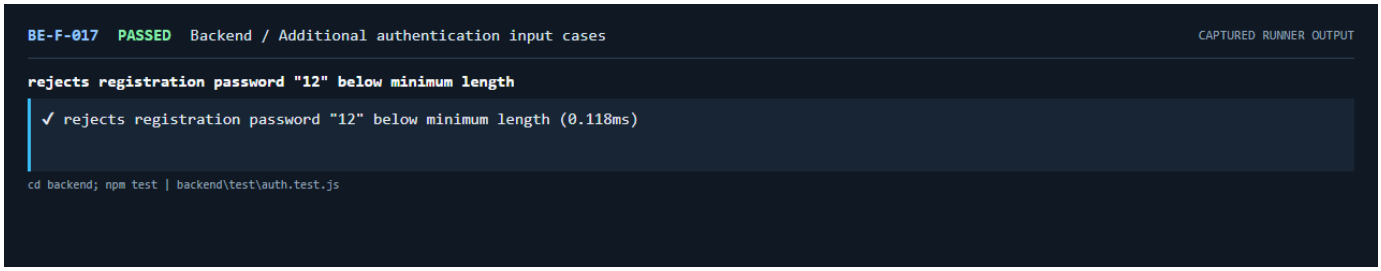
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-016.png.

BE-F-017 - rejects registration password "12" below minimum length

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration password "12" below minimum length.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration password is the boundary value embedded in the case name: "12" below minimum length.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Password boundary is rejected with validation feedback and no auth action.
Actual result	Captured runner result: ✔ rejects registration password "12" below minimum length (0.118ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-017.png.

BE-F-018 - rejects registration password "abc" below minimum length

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration password "abc" below minimum length.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration password is the boundary value embedded in the case name: "abc" below minimum length.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Password boundary is rejected with validation feedback and no auth action.
Actual result	Captured runner result: ✔ rejects registration password "abc" below minimum length (0.0792ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-018.png.

BE-F-019 - rejects registration password "abcd" below minimum length

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration password "abcd" below minimum length.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration password is the boundary value embedded in the case name: "abcd" below minimum length.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Password boundary is rejected with validation feedback and no auth action.
Actual result	Captured runner result: ✔ rejects registration password "abcd" below minimum length (0.0703ms)
Execution status	PASSED

Individual execution evidence



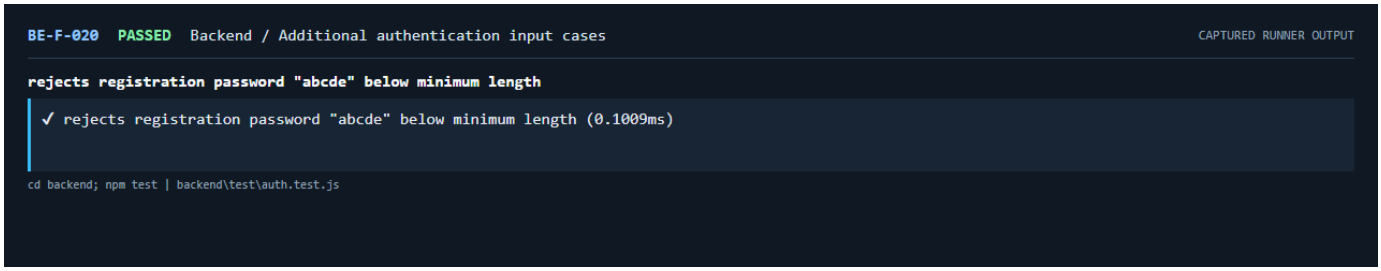
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-019.png.

BE-F-020 - rejects registration password "abcde" below minimum length

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration password "abcde" below minimum length.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration password is the boundary value embedded in the case name: "abcde" below minimum length.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Password boundary is rejected with validation feedback and no auth action.
Actual result	Captured runner result: ✔ rejects registration password "abcde" below minimum length (0.1009ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-020.png.

BE-F-021 - rejects registration password " " below minimum length

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects registration password " " below minimum length.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration password is the boundary value embedded in the case name: " " below minimum length.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Password boundary is rejected with validation feedback and no auth action.
Actual result	Captured runner result: ✔ rejects registration password " " below minimum length (0.0825ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-021.png.

BE-F-022 - rejects unsupported registration role "superuser"

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role "superuser".
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: "superuser".
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role "superuser" (0.1866ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-022.png.

BE-F-023 - rejects unsupported registration role "guest"

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role "guest".
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: "guest".
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role "guest" (0.1314ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-023.png.

BE-F-024 - rejects unsupported registration role "Patient!"

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role "Patient!".
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: "Patient!".
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role "Patient!" (0.1195ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-024.png.

BE-F-025 - rejects unsupported registration role "doctor-admin"

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role "doctor-admin".
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: "doctor-admin".
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role "doctor-admin" (0.1308ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-025.png.

BE-F-026 - rejects unsupported registration role "admin "

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role "admin ".
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: ".
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role "admin " (0.1232ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-026.png.

BE-F-027 - rejects unsupported registration role " PATIENT"

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role " PATIENT".
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: PATIENT".
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role " PATIENT" (0.1144ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-027.png.

BE-F-028 - rejects unsupported registration role 7

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role 7.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: 7.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role 7 (0.1331ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-028.png.

BE-F-029 - rejects unsupported registration role {}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects unsupported registration role {}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Registration role is the unsupported value embedded in the case name: {}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects unsupported registration role {} (0.13ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-029.png.

BE-F-030 - requires both verification fields for {}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: requires both verification fields for {}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Request body: {}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires both verification fields for {} (0.2614ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-030.png.

BE-F-031 - requires both verification fields for {"email":"patient@example.com"}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: requires both verification fields for {"email":"patient@example.com"}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Request body: {"email":"patient@example.com"}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires both verification fields for {"email":"patient@example.com"} (0.1151ms)
Execution status	PASSED

Individual execution evidence



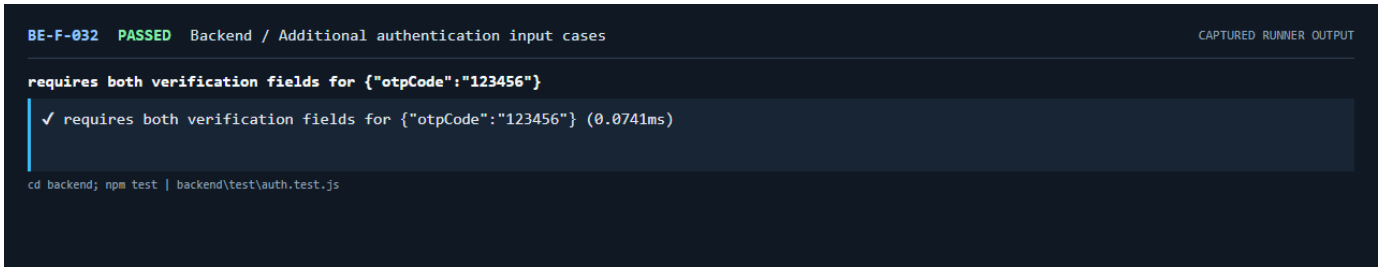
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-031.png.

BE-F-032 - requires both verification fields for {"otpCode":"123456"}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: requires both verification fields for {"otpCode":"123456"}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Request body: {"otpCode":"123456"}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires both verification fields for {"otpCode":"123456"} (0.0741ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-032.png.

BE-F-033 - requires both login credentials for {}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: requires both login credentials for {}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Request body: {}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires both login credentials for {} (0.1378ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-033.png.

BE-F-034 - requires both login credentials for {"email":"patient@example.com"}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: requires both login credentials for {"email":"patient@example.com"}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Request body: {"email":"patient@example.com"}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires both login credentials for {"email":"patient@example.com"} (0.0981ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-034.png.

BE-F-035 - requires both login credentials for {"password":"secret123"}

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: requires both login credentials for {"password":"secret123"}.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Request body: {"password":"secret123"}.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ requires both login credentials for {"password":"secret123"} (0.0663ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-035.png.

BE-F-036 - allows patient access when permitted by patient

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: allows patient access when permitted by patient.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Authorized role reaches next handler exactly once.
Actual result	Captured runner result: ✔ allows patient access when permitted by patient (0.1769ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-036.png.

BE-F-037 - allows doctor access when permitted by doctor

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: allows doctor access when permitted by doctor.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Authorized role reaches next handler exactly once.
Actual result	Captured runner result: ✔ allows doctor access when permitted by doctor (0.0745ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-037.png.

BE-F-038 - allows admin access when permitted by admin

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: allows admin access when permitted by admin.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Authorized role reaches next handler exactly once.
Actual result	Captured runner result: ✔ allows admin access when permitted by admin (0.0611ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-038.png.

BE-F-039 - allows patient access when permitted by doctor, patient

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: allows patient access when permitted by doctor, patient.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Authorized role reaches next handler exactly once.
Actual result	Captured runner result: ✔ allows patient access when permitted by doctor, patient (0.0708ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-039.png.

BE-F-040 - allows doctor access when permitted by admin, doctor

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: allows doctor access when permitted by admin, doctor.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Authorized role reaches next handler exactly once.
Actual result	Captured runner result: ✔ allows doctor access when permitted by admin, doctor (0.0622ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-040.png.

BE-F-041 - allows admin access when permitted by patient, admin

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: allows admin access when permitted by patient, admin.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Authorized role reaches next handler exactly once.
Actual result	Captured runner result: ✔ allows admin access when permitted by patient, admin (0.0801ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-041.png.

BE-F-042 - denies role access when user {"role":"patient"} is not permitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: denies role access when user {"role":"patient"} is not permitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name; includes missing-user case.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Request is rejected with HTTP 403; next handler is not called.
Actual result	Captured runner result: ✔ denies role access when user {"role":"patient"} is not permitted (0.1393ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-042.png.

BE-F-043 - denies role access when user {"role":"doctor"} is not permitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: denies role access when user {"role":"doctor"} is not permitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name; includes missing-user case.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Request is rejected with HTTP 403; next handler is not called.
Actual result	Captured runner result: ✔ denies role access when user {"role":"doctor"} is not permitted (0.1388ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-043.png.

BE-F-044 - denies role access when user {"role":"admin"} is not permitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: denies role access when user {"role":"admin"} is not permitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name; includes missing-user case.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Request is rejected with HTTP 403; next handler is not called.
Actual result	Captured runner result: ✔ denies role access when user {"role":"admin"} is not permitted (0.082ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-044.png.

BE-F-045 - denies role access when user undefined is not permitted

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: denies role access when user undefined is not permitted.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic user role and permitted role list are stated in the test name; includes missing-user case.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Request is rejected with HTTP 403; next handler is not called.
Actual result	Captured runner result: ✔ denies role access when user undefined is not permitted (0.0736ms)
Execution status	PASSED

Individual execution evidence



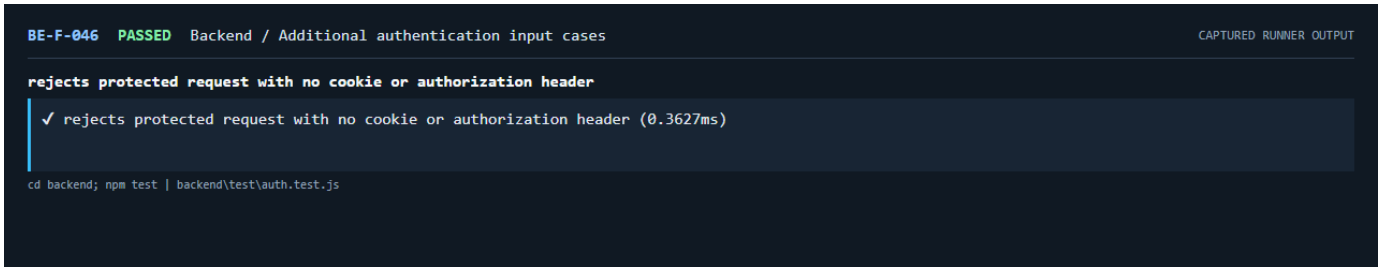
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-045.png.

BE-F-046 - rejects protected request with no cookie or authorization header

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects protected request with no cookie or authorization header.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	JWT/cookie/header scenario: no cookie or authorization header.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects protected request with no cookie or authorization header (0.3627ms)
Execution status	PASSED

Individual execution evidence



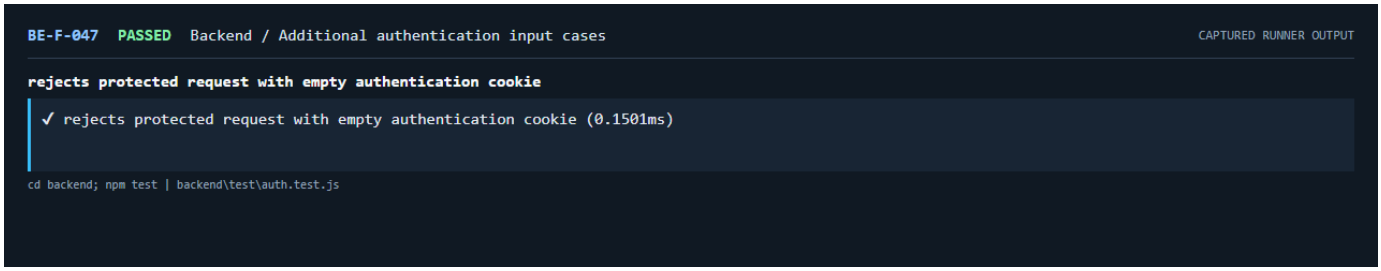
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-046.png.

BE-F-047 - rejects protected request with empty authentication cookie

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects protected request with empty authentication cookie.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	JWT/cookie/header scenario: empty authentication cookie.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: ✔ rejects protected request with empty authentication cookie (0.1501ms)
Execution status	PASSED

Individual execution evidence



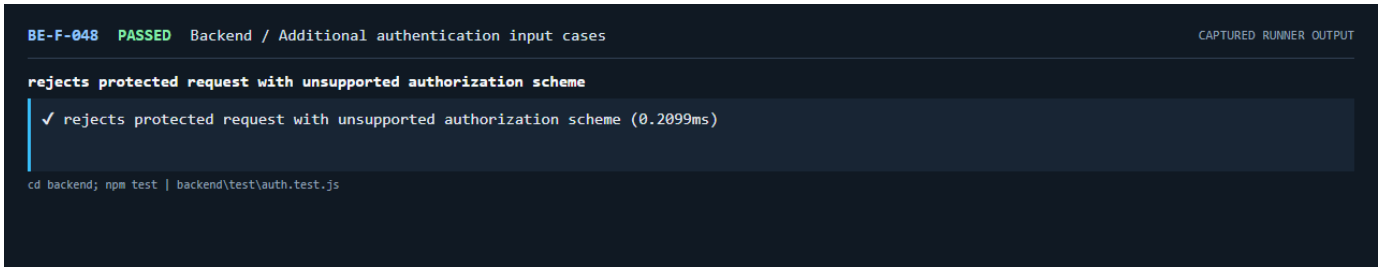
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-047.png.

BE-F-048 - rejects protected request with unsupported authorization scheme

PASSED | Backend | Functional

Feature / module	Additional authentication input cases
Requirement / purpose	Verify the implemented behavior: rejects protected request with unsupported authorization scheme.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	JWT/cookie/header scenario: unsupported authorization scheme.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ rejects protected request with unsupported authorization scheme (0.2099ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-048.png.

BE-F-049 - rejects protected request with malformed token

PASSED | Backend | Functional

Feature / module	JWT authentication checks
Requirement / purpose	Verify the implemented behavior: rejects protected request with malformed token.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	JWT/cookie/header scenario: malformed token.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Protected request is rejected with HTTP 401 and a standardized token-validation error.
Actual result	Captured runner result: ✔ rejects protected request with malformed token (2.0255ms)
Execution status	PASSED

Individual execution evidence



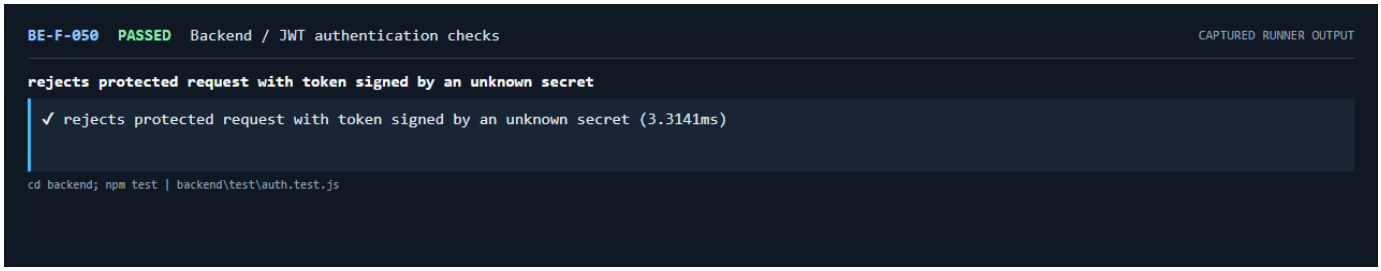
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-049.png.

BE-F-050 - rejects protected request with token signed by an unknown secret

PASSED | Backend | Functional

Feature / module	JWT authentication checks
Requirement / purpose	Verify the implemented behavior: rejects protected request with token signed by an unknown secret.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	JWT/cookie/header scenario: token signed by an unknown secret.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Protected request is rejected with HTTP 401 and a standardized token-validation error.
Actual result	Captured runner result: ✔ rejects protected request with token signed by an unknown secret (3.3141ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-050.png.

BE-F-051 - rejects protected request with expired token

PASSED | Backend | Functional

Feature / module	JWT authentication checks
Requirement / purpose	Verify the implemented behavior: rejects protected request with expired token.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	JWT/cookie/header scenario: expired token.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\auth.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Protected request is rejected with HTTP 401 and a standardized token-validation error.
Actual result	Captured runner result: ✔ rejects protected request with expired token (1.3541ms)
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-F-051.png.

4. Functional testing - AI service

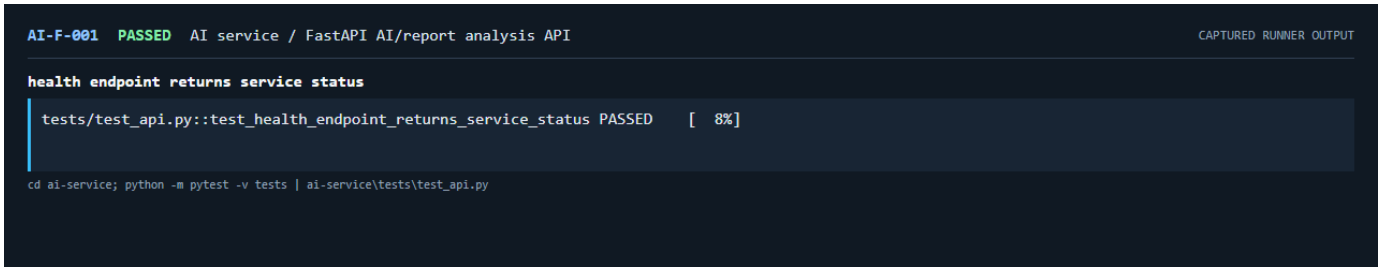
Tests use FastAPI TestClient against actual in-process routes for health, generation, and report analysis. The local LLM boundary is mocked to verify response handling, schema validation, and offline behavior.

AI-F-001 - health endpoint returns service status

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: health endpoint returns service status.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Endpoint/helper returns the asserted status, response shape, and data.
Actual result	Captured runner result: tests/test_api.py::test_health_endpoint_returns_service_status PASSED [8%]
Execution status	PASSED

Individual execution evidence



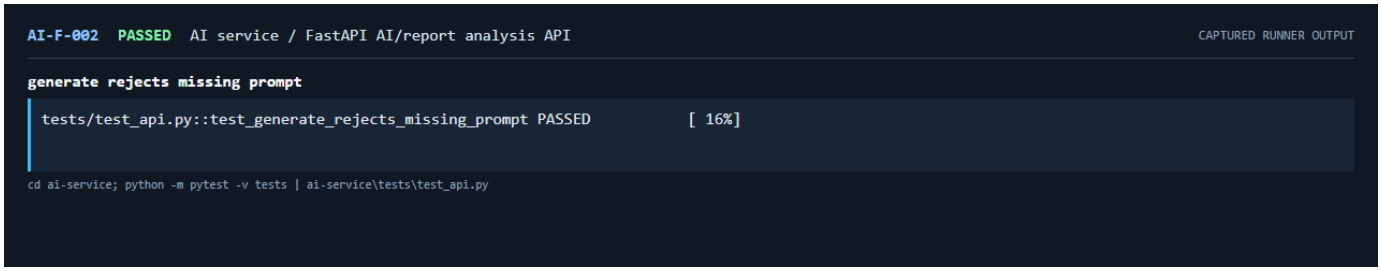
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-001.png.

AI-F-002 - generate rejects missing prompt

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: generate rejects missing prompt.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: tests/test_api.py::test_generate_rejects_missing_prompt PASSED [16%]
Execution status	PASSED

Individual execution evidence



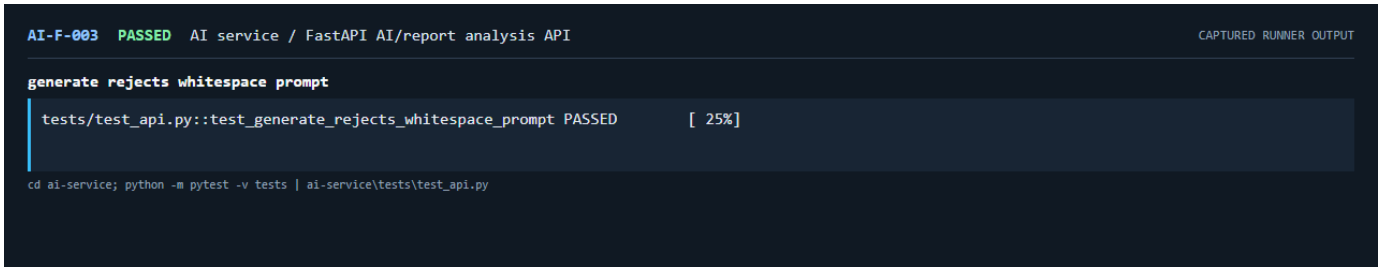
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-002.png.

AI-F-003 - generate rejects whitespace prompt

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: generate rejects whitespace prompt.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: tests/test_api.py::test_generate_rejects_whitespace_prompt PASSED [25%]
Execution status	PASSED

Individual execution evidence



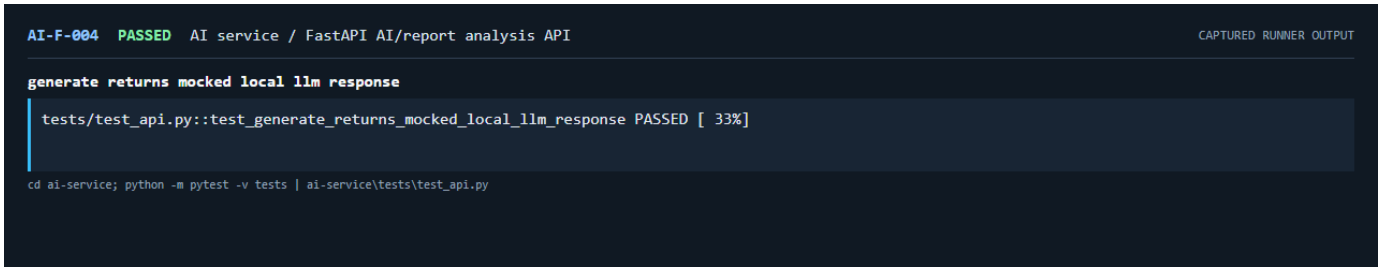
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-003.png.

AI-F-004 - generate returns mocked local llm response

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: generate returns mocked local llm response.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Endpoint/helper returns the asserted status, response shape, and data.
Actual result	Captured runner result: tests/test_api.py::test_generate_returns_mocked_local_llm_response PASSED [33%]
Execution status	PASSED

Individual execution evidence



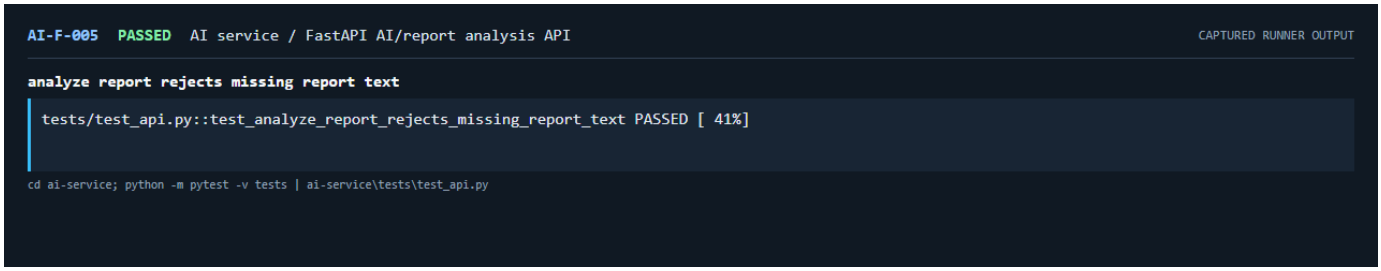
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-004.png.

AI-F-005 - analyze report rejects missing report text

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: analyze report rejects missing report text.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: tests/test_api.py::test_analyze_report_rejects_missing_report_text PASSED [41%]
Execution status	PASSED

Individual execution evidence



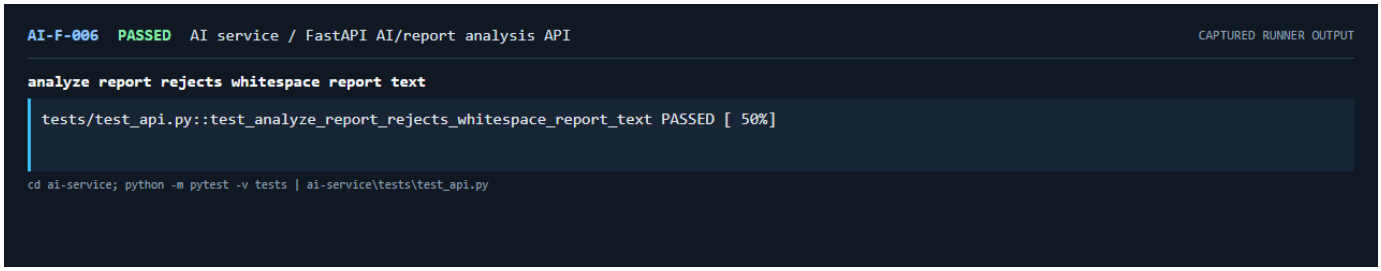
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-005.png.

AI-F-006 - analyze report rejects whitespace report text

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: analyze report rejects whitespace report text.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: tests/test_api.py::test_analyze_report_rejects_whitespace_report_text PASSED [50%]
Execution status	PASSED

Individual execution evidence



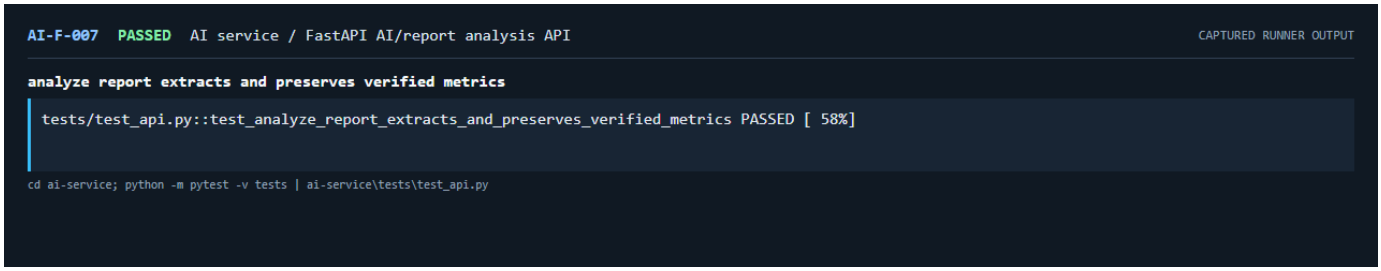
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-006.png.

AI-F-007 - analyze report extracts and preserves verified metrics

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: analyze report extracts and preserves verified metrics.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Analysis response preserves verified value/status, metric coverage, and response schema.
Actual result	Captured runner result: tests/test_api.py::test_analyze_report_extracts_and_preserves_verified_metrics PASSED [58%]
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-007.png.

AI-F-008 - analysis rejects invalid llm schema

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: analysis rejects invalid llm schema.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Invalid model output is rejected with HTTP 422.
Actual result	Captured runner result: tests/test_api.py::test_analysis_rejects_invalid_llm_schema PASSED [66%]
Execution status	PASSED

Individual execution evidence



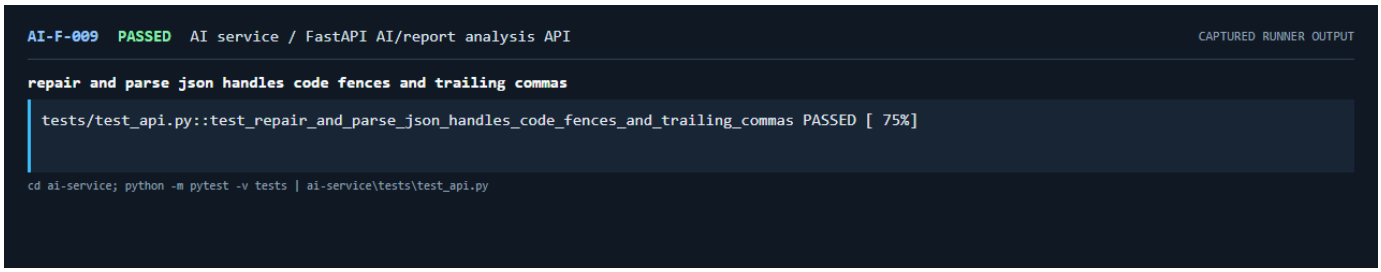
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-008.png.

AI-F-009 - repair and parse json handles code fences and trailing commas

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: repair and parse json handles code fences and trailing commas.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Endpoint/helper returns the asserted status, response shape, and data.
Actual result	Captured runner result: tests/test_api.py::test_repair_and_parse_json_handles_code_fences_and_trailing_commas PASSED [75%]
Execution status	PASSED

Individual execution evidence



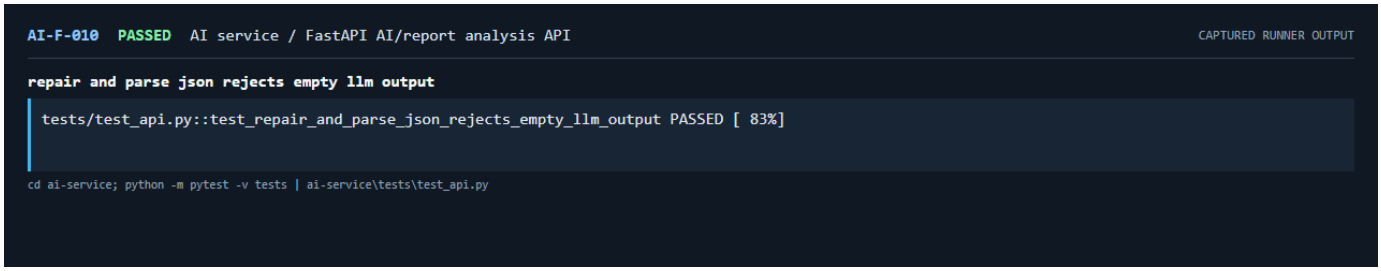
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-009.png.

AI-F-010 - repair and parse json rejects empty llm output

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: repair and parse json rejects empty llm output.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Validation rejects the input with the expected field-specific error or 400/422 response; no success action occurs.
Actual result	Captured runner result: tests/test_api.py::test_repair_and_parse_json_rejects_empty_llm_output PASSED [83%]
Execution status	PASSED

Individual execution evidence



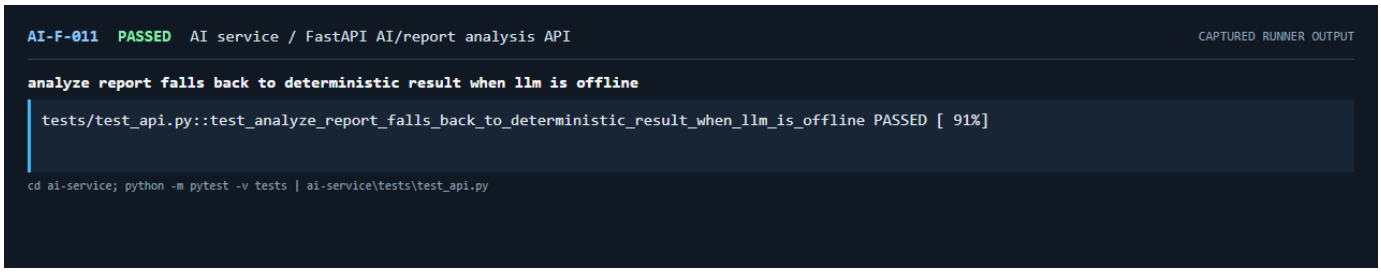
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-010.png.

AI-F-011 - analyze report falls back to deterministic result when llm is offline

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: analyze report falls back to deterministic result when llm is offline.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Service returns the documented fallback or service-unavailable response.
Actual result	Captured runner result: tests/test_api.py::test_analyze_report_falls_back_to_deterministic_result_when_llm_is_offline PASSED [91%]
Execution status	PASSED

Individual execution evidence



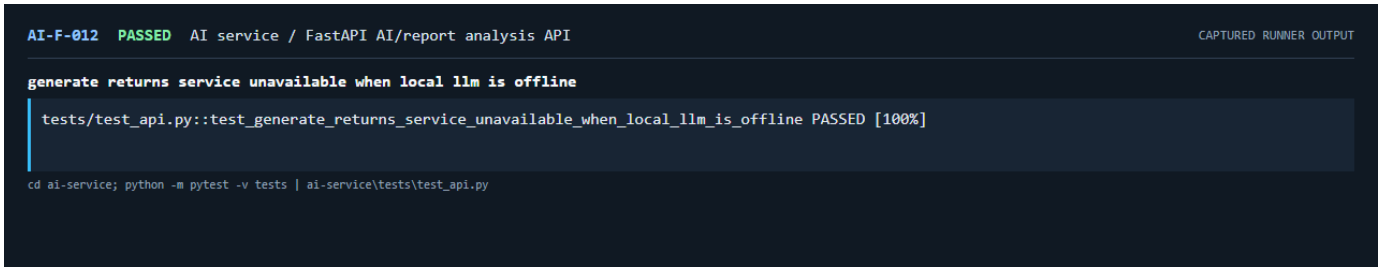
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-011.png.

AI-F-012 - generate returns service unavailable when local llm is offline

PASSED | AI service | Functional

Feature / module	FastAPI AI/report analysis API
Requirement / purpose	Verify the implemented behavior: generate returns service unavailable when local llm is offline.
Preconditions / setup	FastAPI TestClient invokes the actual ASGI route in-process. Local LLM calls are mocked, including explicit offline tests; no external model process or production data is accessed.
Input / test data	Synthetic prompt/report text and mocked LLM response or connection failure as defined in ai-service/tests/test_api.py.
Execution procedure	1. Run `cd ai-service; python -m pytest -v tests`. 2. Execute the test in `ai-service\tests\test_api.py`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Service returns the documented fallback or service-unavailable response.
Actual result	Captured runner result: tests/test_api.py::test_generate_returns_service_unavailable_when_local_llm_is_offline PASSED [100%]
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: AI-F-012.png.

5. Frontend-backend integration

Path	Evidence class	Observed	Status
React login/register -> AuthContext	Mocked component test	Auth function call arguments, form messages, and router result are asserted; no HTTP request reaches Express.	PARTIAL / MOCKED
Axios API client -> adapter	Mocked adapter test	Development base URL, credentials, response unwrapping, and error normalization are asserted.	PARTIAL / MOCKED
Frontend -> Express -> MongoDB	Live integration	No live server/database requests or persistence were executed.	NOT EXECUTED
Express report controller -> FastAPI	Live integration	Network hop from patient API to AI service was not run.	NOT EXECUTED
FastAPI route -> analysis pipeline -> mocked LLM	In-process API	Actual route and extraction/validation pipeline executed with mocked upstream model.	PASSED / MOCKED UPSTREAM
Video frontend -> Socket.IO/WebRTC peer	Live integration	No paired clients, camera, microphone, or signaling session.	NOT EXECUTED

No result in this report should be interpreted as evidence that MongoDB persistence, Cloudinary, Gemini, the live local LLM process, or WebRTC works end-to-end. Those paths require a controlled integration environment and are listed as NOT EXECUTED.

6. Non-functional testing overview

Non-functional checks use measurable criteria only where the current test environment supports them. The controller p95 and React render measurements are in-process test measurements, not public API latency, browser page-load time, or production load results. Axe color-contrast analysis was disabled.

Category	Cases / observed measurement	Result
Accessibility	2 axe runs; 0 enabled-rule violations each (contrast rule disabled).	PASS
Frontend render performance	Login max 3.29ms; button max 0.34ms; asserted <500ms in jsdom.	PASS
Backend local validation performance	p95 0.11ms across 50 in-process calls; asserted <100ms.	PASS
Reliability	Repeated validation/logout/authorization result consistency checks passed.	PASS
Dependency security audit	Frontend: 1 high Axios package finding. Backend: 1 critical, 1 high, 1 moderate vulnerable package.	FAIL
Static analysis	Oxlint exited successfully; 60 warnings.	PASS WITH WARNINGS
Production build	Vite build completed; emitted >500kB chunk advisory.	PASS WITH WARNING
Compatibility/load/DB resource tests	No cross-browser matrix, live-service load, DB performance, or CPU/memory test.	NOT EXECUTED

Non-functional test case index

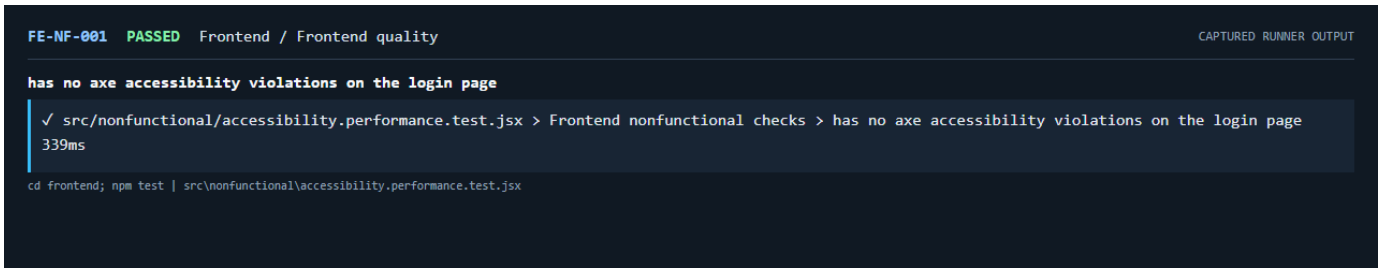
Test Case ID	Test Case Name	Module / short description	Status
FE-NF-001	has no axe accessibility violations on the login page	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-NF-002	has no axe accessibility violations on the registration page	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-NF-003	renders the login page within a 500ms React render budget	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-NF-004	renders and updates an enabled button without exceeding the render budget	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
BE-NF-001	keeps registration validation p95 below 100ms across 50 requests	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
BE-NF-002	returns the same validation response across 50 repeated requests	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
BE-NF-003	clears the session cookie consistently across 25 logouts	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
BE-NF-004	returns consistent forbidden responses across 50 unauthorized role checks	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
FE-NF-SEC-01	Frontend npm production dependency audit	Production dependency security audit frontend\package-lock.json	FAILED
BE-NF-SEC-01	Backend npm production dependency audit	Production dependency security audit backend\package-lock.json	FAILED
FE-NF-QA-01	Frontend Oxlint static analysis	Static analysis / maintainability frontend\src\ (workspace)	PASSED
FE-NF-QA-02	Frontend production build	Build / release quality frontend\vite.config.js	PASSED

FE-NF-001 - has no axe accessibility violations on the login page

PASSED | Frontend | Non-functional

Feature / module	Frontend quality
Requirement / purpose	Verify the implemented behavior: has no axe accessibility violations on the login page.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\nonfunctional\accessibility.performance.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Axe reports zero enabled-rule accessibility violations for the rendered page.
Actual result	Captured runner result: ✓ src/nonfunctional/accessibility.performance.test.jsx > Frontend nonfunctional checks > has no axe accessibility violations on the login page 339ms
Execution status	PASSED

Individual execution evidence



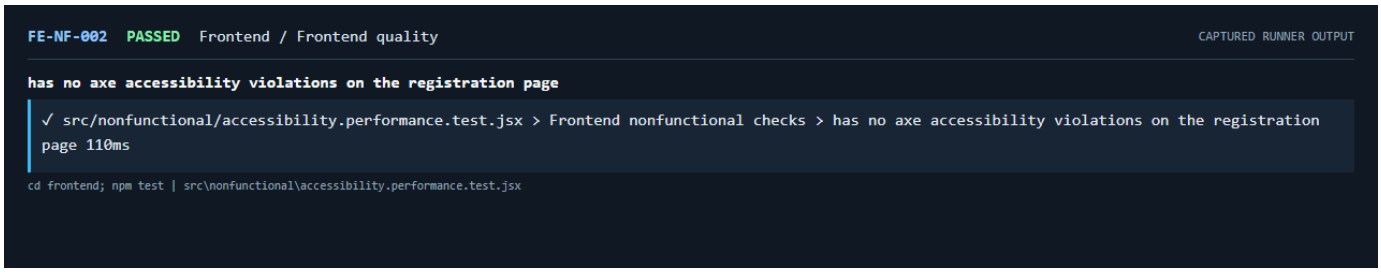
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-001.png.

FE-NF-002 - has no axe accessibility violations on the registration page

PASSED | Frontend | Non-functional

Feature / module	Frontend quality
Requirement / purpose	Verify the implemented behavior: has no axe accessibility violations on the registration page.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\nonfunctional\accessibility.performance.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Axe reports zero enabled-rule accessibility violations for the rendered page.
Actual result	Captured runner result: ✓ src/nonfunctional/accessibility.performance.test.jsx > Frontend nonfunctional checks > has no axe accessibility violations on the registration page 110ms
Execution status	PASSED

Individual execution evidence



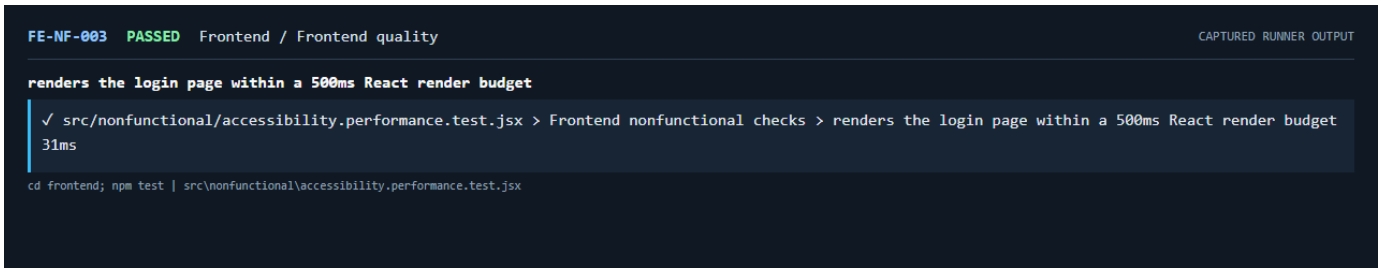
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-002.png.

FE-NF-003 - renders the login page within a 500ms React render budget

PASSED | Frontend | Non-functional

Feature / module	Frontend quality
Requirement / purpose	Verify the implemented behavior: renders the login page within a 500ms React render budget.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\nonfunctional\accessibility.performance.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/nonfunctional/accessibility.performance.test.jsx > Frontend nonfunctional checks > renders the login page within a 500ms React render budget 31ms
Execution status	PASSED

Individual execution evidence



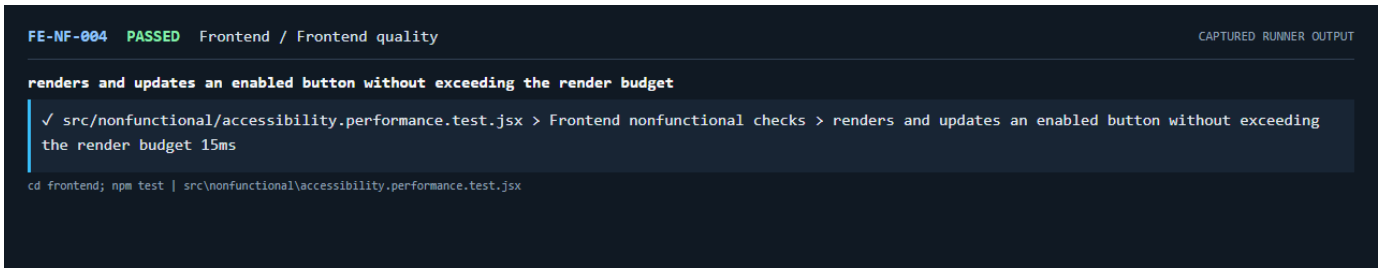
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-003.png.

FE-NF-004 - renders and updates an enabled button without exceeding the render budget

PASSED | Frontend | Non-functional

Feature / module	Frontend quality
Requirement / purpose	Verify the implemented behavior: renders and updates an enabled button without exceeding the render budget.
Preconditions / setup	Rendered/exercised with Vitest, React Testing Library, jsdom, and MemoryRouter where relevant. Authentication actions or Axios adapters are mocked in these component/service tests; this is not a live backend integration run.
Input / test data	Synthetic values defined by this test; no production or patient data.
Execution procedure	1. Run `cd frontend; npm test`. 2. Execute the test in `src\nonfunctional\accessibility.performance.test.jsx`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✓ src/nonfunctional/accessibility.performance.test.jsx > Frontend nonfunctional checks > renders and updates an enabled button without exceeding the render budget 15ms
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-004.png.

BE-NF-001 - keeps registration validation p95 below 100ms across 50 requests

PASSED | Backend | Non-functional

Feature / module	Backend reliability/performance
Requirement / purpose	Verify the implemented behavior: keeps registration validation p95 below 100ms across 50 requests.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\nonfunctional.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Measured duration is below the explicitly asserted 500ms UI or 100ms controller threshold.
Actual result	Captured runner result: ✔ keeps registration validation p95 below 100ms across 50 requests (6.2637ms) Measurement: [MEASURE] Registration validation p95 across 50 in-process calls: 0.11ms (limit <100ms).
Execution status	PASSED

Individual execution evidence



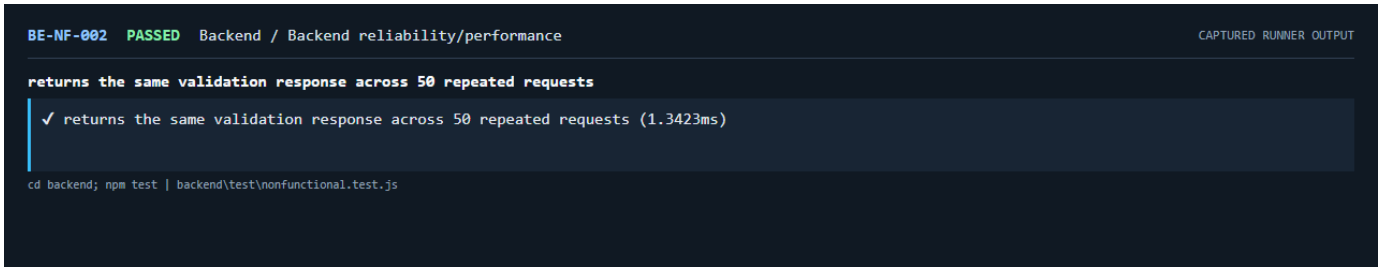
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-NF-001.png.

BE-NF-002 - returns the same validation response across 50 repeated requests

PASSED | Backend | Non-functional

Feature / module	Backend reliability/performance
Requirement / purpose	Verify the implemented behavior: returns the same validation response across 50 repeated requests.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\nonfunctional.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The behavior and response asserted by the named test are satisfied.
Actual result	Captured runner result: ✔ returns the same validation response across 50 repeated requests (1.3423ms)
Execution status	PASSED

Individual execution evidence



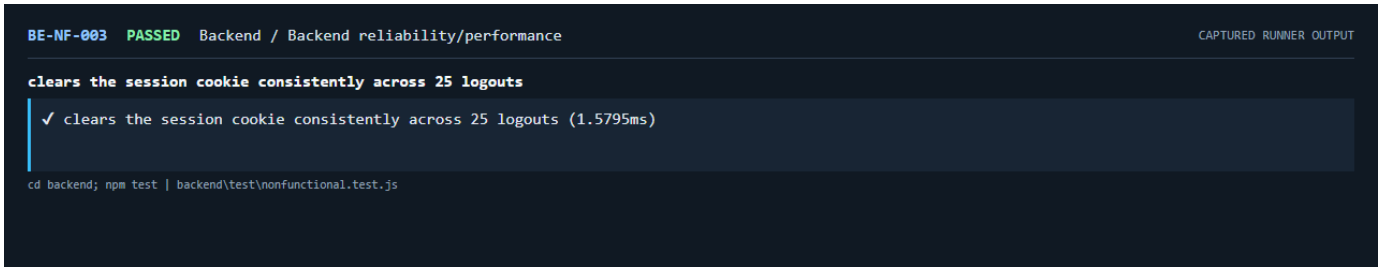
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-NF-002.png.

BE-NF-003 - clears the session cookie consistently across 25 logouts

PASSED | Backend | Non-functional

Feature / module	Backend reliability/performance
Requirement / purpose	Verify the implemented behavior: clears the session cookie consistently across 25 logouts.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Mock response object; no browser cookie jar or external service.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\nonfunctional.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Response confirms logout and clears the HTTP-only JWT cookie.
Actual result	Captured runner result: ✔ clears the session cookie consistently across 25 logouts (1.5795ms)
Execution status	PASSED

Individual execution evidence



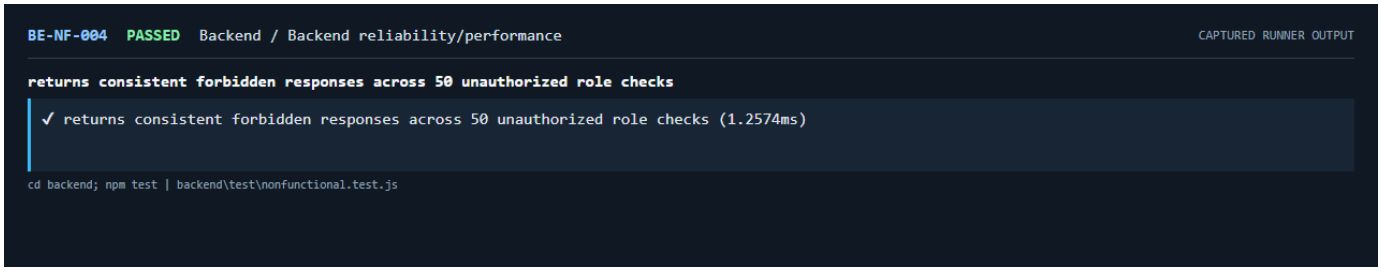
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-NF-003.png.

BE-NF-004 - returns consistent forbidden responses across 50 unauthorized role checks

PASSED | Backend | Non-functional

Feature / module	Backend reliability/performance
Requirement / purpose	Verify the implemented behavior: returns consistent forbidden responses across 50 unauthorized role checks.
Preconditions / setup	Calls the actual Node.js controller or middleware function with synthetic requests and a mocked Express response. No HTTP server, MongoDB query, or persisted record is involved.
Input / test data	Synthetic Express controller/middleware request body defined in the test source.
Execution procedure	1. Run `cd backend; npm test`. 2. Execute the test in `backend\test\nonfunctional.test.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Request is rejected with HTTP 403; next handler is not called.
Actual result	Captured runner result: ✔ returns consistent forbidden responses across 50 unauthorized role checks (1.2574ms)
Execution status	PASSED

Individual execution evidence



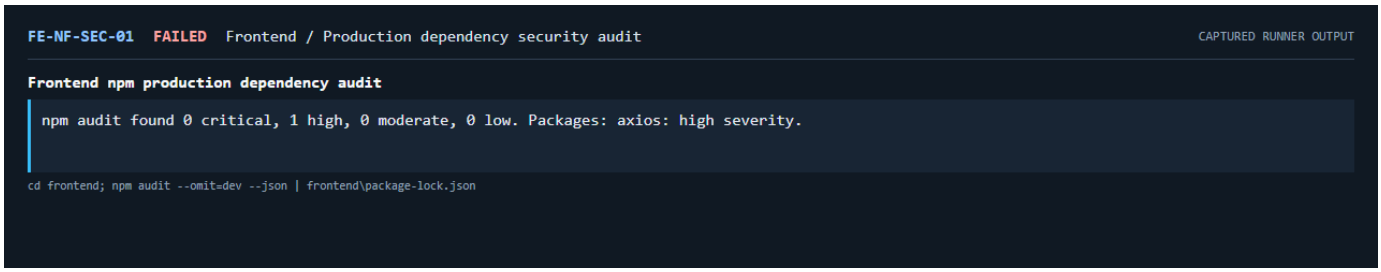
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-NF-004.png.

FE-NF-SEC-01 - Frontend npm production dependency audit

FAILED | Frontend | Non-functional

Feature / module	Production dependency security audit
Requirement / purpose	Verify the implemented behavior: Frontend npm production dependency audit.
Preconditions / setup	npm audit checked the committed production dependency tree against the advisory database at execution time. This is a dependency advisory scan, not a penetration test or proof of exploitability.
Input / test data	Locked production dependency graph; development dependencies excluded.
Execution procedure	1. Run `cd frontend; npm audit --omit=dev --json`. 2. Execute the test in `frontend\package-lock.json`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	No known production dependency advisories should be reported by npm audit.
Actual result	Captured runner result: npm audit found 0 critical, 1 high, 0 moderate, 0 low. Packages: axios: high severity. This is an observed audit finding, not a test-runner assertion failure.
Execution status	FAILED

Individual execution evidence



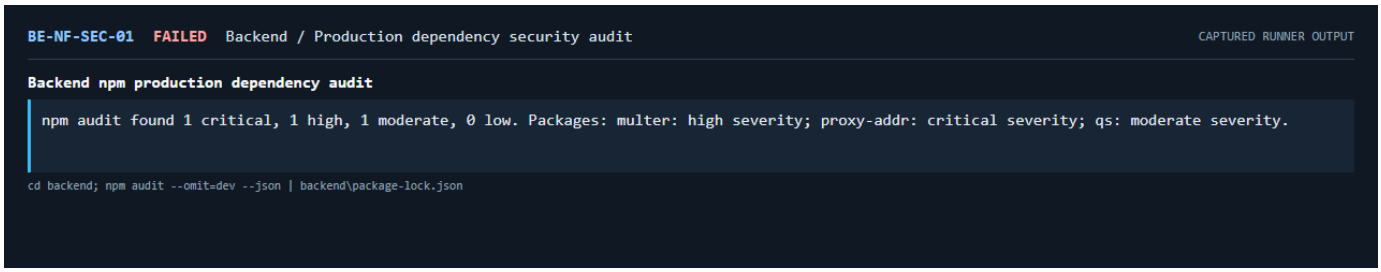
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-SEC-01.png.

BE-NF-SEC-01 - Backend npm production dependency audit

FAILED | Backend | Non-functional

Feature / module	Production dependency security audit
Requirement / purpose	Verify the implemented behavior: Backend npm production dependency audit.
Preconditions / setup	npm audit checked the committed production dependency tree against the advisory database at execution time. This is a dependency advisory scan, not a penetration test or proof of exploitability.
Input / test data	Locked production dependency graph; development dependencies excluded.
Execution procedure	1. Run `cd backend; npm audit --omit=dev --json`. 2. Execute the test in `backend\package-lock.json`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	No known production dependency advisories should be reported by npm audit.
Actual result	Captured runner result: npm audit found 1 critical, 1 high, 1 moderate, 0 low. Packages: multer: high severity; proxy-addr: critical severity; qs: moderate severity. This is an observed audit finding, not a test-runner assertion failure.
Execution status	FAILED

Individual execution evidence



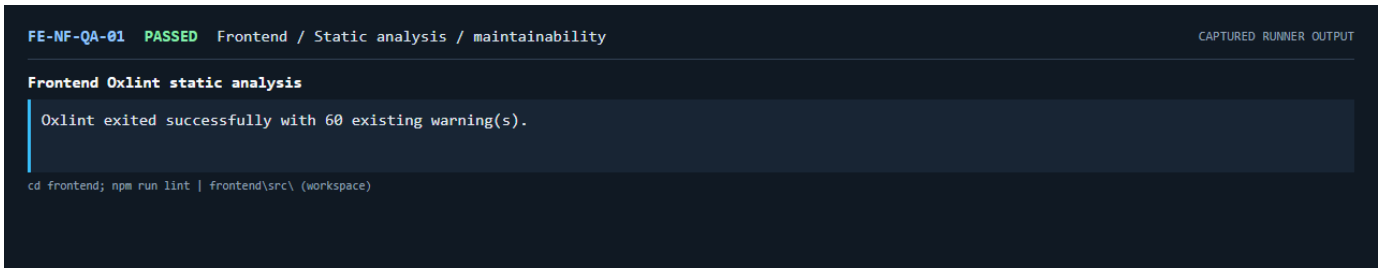
Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: BE-NF-SEC-01.png.

FE-NF-QA-01 - Frontend Oxlint static analysis

PASSED | Frontend | Non-functional

Feature / module	Static analysis / maintainability
Requirement / purpose	Verify the implemented behavior: Frontend Oxlint static analysis.
Preconditions / setup	Runs the repository's configured Oxlint command over the frontend source.
Input / test data	All source files included by the default Oxlint invocation.
Execution procedure	1. Run `cd frontend; npm run lint`. 2. Execute the test in `frontend\src\ (workspace)`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Oxlint completes with no process error; warnings are separately reported.
Actual result	Captured runner result: Oxlint exited successfully with 60 existing warning(s).
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-QA-01.png.

FE-NF-QA-02 - Frontend production build

PASSED | Frontend | Non-functional

Feature / module	Build / release quality
Requirement / purpose	Verify the implemented behavior: Frontend production build.
Preconditions / setup	Runs the configured Vite production build. Build completion is successful; the captured output also reports a JavaScript chunk larger than 500 kB.
Input / test data	Complete Vite production bundle from current frontend source.
Execution procedure	1. Run `cd frontend; npm run build`. 2. Execute the test in `frontend\vite.config.js`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	Vite completes a production build successfully.
Actual result	Captured runner result: ■[32m✓ built in 2.10s■[39m
Execution status	PASSED

Individual execution evidence



Chromium screenshot of a local evidence page containing the exact captured runner/audit output excerpt shown in the adjacent raw evidence log. Source log: test-report-evidence/logs; PNG: FE-NF-QA-02.png.

7. Not executed: coverage gaps and required environment

The following project-relevant scenarios are intentionally marked NOT EXECUTED. They are not counted as passing tests, and no screenshots are fabricated. Each row states the limitation and a suitable execution procedure for a future isolated staging run.

Not-executed scenario index

Test Case ID	Test Case Name	Module / short description	Status
INT-01	Patient registration, OTP, login and logout against live Express + MongoDB	Integration / Authentication Not in executed test files	NOT EXECUTED
DB-01	Patient health profile create/read/update persistence	Database / Patient health Not in executed test files	NOT EXECUTED
DB-02	Medical record upload/list/delete with storage integration	Database / Records and upload Not in executed test files	NOT EXECUTED
DB-03	Medication and appointment CRUD persistence	Database / Patient scheduling Not in executed test files	NOT EXECUTED
DB-04	Doctor access consent and role-protected patient data	Database / Authorization Not in executed test files	NOT EXECUTED
DB-05	Admin user status/role changes and audit-log persistence	Database / Administration Not in executed test files	NOT EXECUTED
DB-06	Community posts, comments, and likes persistence	Database / Community Not in executed test files	NOT EXECUTED
DB-07	Notification read/delete state persistence	Database / Notifications Not in executed test files	NOT EXECUTED
DB-08	Care-plan task completion persistence	Database / Care plans Not in executed test files	NOT EXECUTED
INT-02	Frontend patient report analyzer through Express to FastAPI	Integration / Report analysis Not in executed test files	NOT EXECUTED
EXT-01	Real local LLM generation and analysis response	External service / AI Not in executed test files	NOT EXECUTED
EXT-02	Emergency assistant response through Gemini	External service / Emergency assistant Not in executed test files	NOT EXECUTED
EXT-03	Cloudinary upload and remote deletion	External service / File storage Not in executed test files	NOT EXECUTED
INT-03	Two-peer WebRTC/Socket.IO consultation workflow	Integration / Video consultation Not in executed test files	NOT EXECUTED
NF-01	Live Express endpoint response-time and concurrent-user load test	Non-functional / Performance Not in executed test files	NOT EXECUTED
NF-02	Browser page-load and mobile/tablet responsiveness across supported browsers	Non-functional / Compatibility Not in executed test files	NOT EXECUTED
NF-03	Keyboard-only, screen-reader, and full WCAG color-contrast audit	Non-functional / Accessibility Not in executed test files	NOT EXECUTED
NF-04	MongoDB outage recovery, data-volume scalability, and process resource usage	Non-functional / Reliability and scalability Not in executed test files	NOT EXECUTED
SEC-01	Dynamic injection, production configuration, secret exposure, and route authorization audit	Non-functional / Security Not in executed test files	NOT EXECUTED

INT-01 - Patient registration, OTP, login and logout against live Express + MongoDB

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Integration / Authentication
Requirement / purpose	Verify the implemented behavior: Patient registration, OTP, login and logout against live Express + MongoDB.
Preconditions / setup	Run isolated test database; register user, fetch OTP via dev-only endpoint, verify, log in, call protected /me, log out; assert persisted states and cookie/token behavior.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - No live Express/MongoDB environment was configured for this run; controller unit tests do not persist accounts.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-01 - Patient health profile create/read/update persistence

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Patient health
Requirement / purpose	Verify the implemented behavior: Patient health profile create/read/update persistence.
Preconditions / setup	Against disposable MongoDB, POST health data, GET it, update values, and verify persisted representation.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - MongoDB connection and isolated test dataset were not exercised.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-02 - Medical record upload/list/delete with storage integration

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Records and upload
Requirement / purpose	Verify the implemented behavior: Medical record upload/list/delete with storage integration.
Preconditions / setup	Upload an allowed test document, list record, verify metadata/storage reference, delete, and verify removal.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Requires MongoDB plus upload storage or Cloudinary credentials; neither integration was run.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-03 - Medication and appointment CRUD persistence

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Patient scheduling
Requirement / purpose	Verify the implemented behavior: Medication and appointment CRUD persistence.
Preconditions / setup	Create synthetic medication and appointment, read by patient, modify/cancel, and verify database state.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - No real database-backed create/read/update/delete test executed.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-04 - Doctor access consent and role-protected patient data

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Authorization
Requirement / purpose	Verify the implemented behavior: Doctor access consent and role-protected patient data.
Preconditions / setup	Attempt access before consent, grant access, read permitted patient data, revoke access, then verify denial.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Requires persisted patient/doctor users and MongoDB relationships.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-05 - Admin user status/role changes and audit-log persistence

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Administration
Requirement / purpose	Verify the implemented behavior: Admin user status/role changes and audit-log persistence.
Preconditions / setup	Change a synthetic user's status/role and doctor verification; confirm the audit entry and updated database state.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Requires seeded admin and users in an isolated MongoDB database.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-06 - Community posts, comments, and likes persistence

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Community
Requirement / purpose	Verify the implemented behavior: Community posts, comments, and likes persistence.
Preconditions / setup	Create a synthetic post, read feed, comment, toggle like, and verify persistence and ownership rules.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Post controller/model operations were not exercised by current test files.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-07 - Notification read/delete state persistence

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Notifications
Requirement / purpose	Verify the implemented behavior: Notification read/delete state persistence.
Preconditions / setup	Create/seed notifications, mark one/all read, delete one, and verify stored state.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Requires persisted notification records and a live backend.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

DB-08 - Care-plan task completion persistence

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Database / Care plans
Requirement / purpose	Verify the implemented behavior: Care-plan task completion persistence.
Preconditions / setup	Toggle a task for an authorized patient/doctor and verify state survives a subsequent read.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - No live MongoDB care-plan records were available.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

INT-02 - Frontend patient report analyzer through Express to FastAPI

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Integration / Report analysis
Requirement / purpose	Verify the implemented behavior: Frontend patient report analyzer through Express to FastAPI.
Preconditions / setup	Start Express + FastAPI; submit UI report; verify request payload, service response, displayed metrics, and stored record.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - React service tests use mocked adapters; backend report controller and network hop were not exercised.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

EXT-01 - Real local LLM generation and analysis response

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	External service / AI
Requirement / purpose	Verify the implemented behavior: Real local LLM generation and analysis response.
Preconditions / setup	Run supported model service and exercise /generate and /analyze-report using non-sensitive synthetic test text.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Local llama.cpp endpoint 127.0.0.1:8081 was mocked and was not verified live.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

EXT-02 - Emergency assistant response through Gemini

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	External service / Emergency assistant
Requirement / purpose	Verify the implemented behavior: Emergency assistant response through Gemini.
Preconditions / setup	Use a non-sensitive synthetic emergency scenario with a test key; verify response validation, errors, and rate/usage controls.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Requires Gemini credentials/network; external call was not invoked.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

EXT-03 - Cloudinary upload and remote deletion

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	External service / File storage
Requirement / purpose	Verify the implemented behavior: Cloudinary upload and remote deletion.
Preconditions / setup	Use a dedicated test account and non-sensitive test file; verify upload URL, cleanup, and failure handling.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - No Cloudinary credentials or remote upload/deletion run.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

INT-03 - Two-peer WebRTC/Socket.IO consultation workflow

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Integration / Video consultation
Requirement / purpose	Verify the implemented behavior: Two-peer WebRTC/Socket.IO consultation workflow.
Preconditions / setup	Connect two test clients, join room, exchange offer/answer/ICE, send chat, and verify disconnect cleanup.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - No paired browser peers, live Socket.IO server, camera/microphone, or signaling session used.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

NF-01 - Live Express endpoint response-time and concurrent-user load test

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Non-functional / Performance
Requirement / purpose	Verify the implemented behavior: Live Express endpoint response-time and concurrent-user load test.
Preconditions / setup	Run isolated service and database with defined concurrency, warm-up, p95/p99 thresholds, and record CPU/memory.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Measured backend p95 is controller-only in-process validation (0.11ms), not HTTP/database service latency; no load harness used.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

NF-02 - Browser page-load and mobile/tablet responsiveness across supported browsers

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Non-functional / Compatibility
Requirement / purpose	Verify the implemented behavior: Browser page-load and mobile/tablet responsiveness across supported browsers.
Preconditions / setup	Test supported browser versions at desktop/tablet/mobile viewports and record page errors, overflow, and key journeys.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Vitest uses jsdom; no cross-browser or multi-viewport browser matrix was executed.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

NF-03 - Keyboard-only, screen-reader, and full WCAG color-contrast audit

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Non-functional / Accessibility
Requirement / purpose	Verify the implemented behavior: Keyboard-only, screen-reader, and full WCAG color-contrast audit.
Preconditions / setup	Run keyboard navigation, screen reader checks, and a browser-based WCAG contrast audit across core routes.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Axe checks on login/register passed but explicitly disabled color-contrast; no screen reader or keyboard-only full-flow audit was run.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

NF-04 - MongoDB outage recovery, data-volume scalability, and process resource usage

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Non-functional / Reliability and scalability
Requirement / purpose	Verify the implemented behavior: MongoDB outage recovery, data-volume scalability, and process resource usage.
Preconditions / setup	Use a disposable database, inject connection loss/recovery, increase seeded data and request volume, and capture resource metrics.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - No MongoDB outage, large dataset, sustained workload, or CPU/memory telemetry was available.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

SEC-01 - Dynamic injection, production configuration, secret exposure, and route authorization audit

NOT EXECUTED | Cross-service / Environment | Not executed

Feature / module	Non-functional / Security
Requirement / purpose	Verify the implemented behavior: Dynamic injection, production configuration, secret exposure, and route authorization audit.
Preconditions / setup	Use an authorized staging environment; verify input handling, route controls, secret redaction, cookie flags, and security headers.
Input / test data	Would use isolated synthetic records only; no real patient data.
Execution procedure	1. Run `Not run`. 2. Execute the test in `Not in executed test files`. 3. Apply the test's recorded inputs. 4. Evaluate the assertions reported by the runner.
Expected result	The scenario should meet its stated acceptance criteria when run in the required isolated environment.
Actual result	NOT EXECUTED - Only JWT rejection/role middleware unit checks and npm dependency audits executed; no dynamic penetration test or production secret/config review performed.
Execution status	NOT EXECUTED

Evidence screenshot: NOT AVAILABLE - this scenario was not executed. No screenshot or successful outcome has been fabricated.

8. Failed checks, defects, and recommendations

Observed security audit findings

- Frontend production audit: axios is reported HIGH; locked range is below 1.20.0. npm audit reported fix available. Review the current supported patched release and validate compatibility before updating.

- Backend production audit: proxy-addr is reported CRITICAL (affected range 1.1.0-2.0.7); multer HIGH (package entry reports multiple multipart DoS advisories); qs MODERATE. npm audit reports fixes available. Prioritize a compatible dependency update and rerun tests/audit.

- The dependency audit is advisory matching, not proof that a vulnerable code path is exploitable in this deployment. No exploit validation was performed.

Quality observations and resolved test-discovered issues

- Final automated suites: 122/122 passed; no final test case failed or was skipped.

- Oxlint: 60 warnings (unused imports/variables/parameters) across frontend source; exit code was successful. Warnings remain for normal code-quality cleanup.

- Production build: Vite succeeded, with a bundle chunk larger than the configured 500kB warning threshold.

- Test work added semantic labels to shared inputs and the registration role select, and guarded non-string registration roles with a 400 response. Those behaviors are included in the final passing run.

- No test coverage percentage is reported: no coverage run was performed. No live database CRUD, real Express API, real upstream LLM, Cloudinary, Gemini, or WebRTC integration was verified.

Recommended next steps

- Update vulnerable frontend/backend production dependencies in a separate reviewed change; rerun npm audit and all functional suites.

- Add isolated MongoDB integration tests for representative patient/doctor/admin CRUD and role boundaries.

- Add end-to-end API tests for frontend -> Express and Express -> AI service using non-sensitive seeded data.

- Run browser-based axe/contrast checks and keyboard-only workflows at desktop and mobile viewports.

- Run HTTP-level concurrency/load tests with defined p95/p99 thresholds and collect DB/CPU/memory telemetry.

9. Final execution conclusion

The project's executed unit/component/in-process API checks passed: 55 frontend, 55 backend, and 12 AI-service tests. Additional React render, accessibility, repeated-response, controller latency, lint, and build checks completed. Two npm production dependency audits failed their zero-advisory acceptance expectation and identified findings requiring remediation. The report does not claim live database persistence, external service, full end-to-end integration, cross-browser compatibility, or production-load validation.

Raw evidence files

- test-report-evidence/logs/frontend-tests.log
- test-report-evidence/logs/backend-tests.log
- test-report-evidence/logs/ai-service-tests.log
- test-report-evidence/logs/frontend-lint.log
- test-report-evidence/logs/frontend-build.log
- test-report-evidence/logs/frontend-audit.json
- test-report-evidence/logs/backend-audit.json
- test-report-evidence/screenshots/ (one PNG per executed report case)
- scripts/generate_test_report.py

Appendix: executed test case index

Automated and quality check index

Test Case ID	Test Case Name	Module / short description	Status
FE-F-001	renders an input with type text	Shared form input src\components\Input.test.jsx	PASSED
FE-F-002	renders an input with type email	Shared form input src\components\Input.test.jsx	PASSED
FE-F-003	renders an input with type password	Shared form input src\components\Input.test.jsx	PASSED
FE-F-004	renders an input with type number	Shared form input src\components\Input.test.jsx	PASSED
FE-F-005	renders an input with type tel	Shared form input src\components\Input.test.jsx	PASSED
FE-F-006	associates a provided id with its visible label	Shared form input src\components\Input.test.jsx	PASSED
FE-F-007	shows the validation error and error border style	Shared form input src\components\Input.test.jsx	PASSED
FE-F-008	forwards placeholder, value, and change events	Shared form input src\components\Input.test.jsx	PASSED
FE-F-009	builds a development API URL and includes credentials	Axios API client src\services\api.test.js	PASSED
FE-F-010	unwraps the standard successful API response body	Axios API client src\services\api.test.js	PASSED
FE-F-011	normalizes API errors using the server error message	Axios API client src\services\api.test.js	PASSED
FE-F-012	renders the primary variant styles	Shared action button src\components\Button.test.jsx	PASSED
FE-F-013	renders the secondary variant styles	Shared action button src\components\Button.test.jsx	PASSED
FE-F-014	renders the danger variant styles	Shared action button src\components\Button.test.jsx	PASSED
FE-F-015	disables the button when disabled is true	Shared action button src\components\Button.test.jsx	PASSED
FE-F-016	shows a loading indicator and disables the button while loading	Shared action button src\components\Button.test.jsx	PASSED
FE-F-017	appends a caller-provided class name	Shared action button src\components\Button.test.jsx	PASSED
FE-F-018	forwards button type and click handlers	Shared action button src\components\Button.test.jsx	PASSED
FE-F-019	keeps enabled buttons interactive	Shared action button src\components\Button.test.jsx	PASSED
FE-F-020	renders the email and password fields	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-021	requires a name, email, and password	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-022	shows required-field errors without calling the login action	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-NF-001	has no axe accessibility violations on the login page	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-NF-002	has no axe accessibility violations on the registration page	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-NF-003	renders the login page within a 500ms React render budget	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-NF-004	renders and updates an enabled button without exceeding the render budget	Frontend quality src\nonfunctional\accessibility.performance.test.jsx	PASSED
FE-F-023	shows a validation error for malformed email addresses	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-024	rejects passwords shorter than six characters	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-025	logs in and redirects to the dashboard after valid submission	Authentication / Login src\pages\Login.test.jsx	PASSED

Test Case ID	Test Case Name	Module / short description	Status
FE-F-026	registers valid details and routes to OTP verification	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-027	offers OTP verification when the backend rejects an unverified account	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-028	rejects malformed email address invalid	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-029	rejects malformed email address invalid	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-030	rejects malformed email address missing-at.example.com	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-031	rejects malformed email address user@	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-032	rejects malformed email address missing-at.example.com	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-033	rejects malformed email address @example.com	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-034	rejects malformed email address user@	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-035	rejects malformed email address user@example	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-036	treats whitespace-only email as missing	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-037	requires a password when the email is valid	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-038	clears the email validation message when the email is edited	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-039	rejects malformed email address @example.com	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-040	rejects malformed email address user@example	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-041	rejects a whitespace-only name	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-042	requires a password when name and email are valid	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-043	shows the backend error message when login fails for another reason	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-044	shows the verified confirmation banner from the query string	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-045	clears the name validation message when the name is edited	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-046	returns to the originally requested route after login	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-047	shows the backend registration error	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-048	submits email addresses containing plus tags unchanged	Authentication / Login src\pages\Login.test.jsx	PASSED
FE-F-049	submits the selected doctor role	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-050	submits the selected admin role	Authentication / Registration src\pages\Register.test.jsx	PASSED
FE-F-051	URL-encodes plus signs in the email when navigating to OTP verification	Authentication / Registration src\pages\Register.test.jsx	PASSED
BE-F-001	rejects registration when required fields are missing	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-002	rejects registration passwords shorter than six characters	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-003	rejects registration with an unsupported role	Authentication endpoints backend\test\auth.test.js	PASSED

Test Case ID	Test Case Name	Module / short description	Status
BE-F-004	requires email and OTP code for verification	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-005	requires credentials for login	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-006	clears the authentication cookie on logout	Authentication endpoints backend\test\auth.test.js	PASSED
BE-F-007	allows requests matching an authorized role	Role authorization middleware backend\test\auth.test.js	PASSED
BE-F-008	rejects requests with an unauthorized role	Role authorization middleware backend\test\auth.test.js	PASSED
BE-F-009	rejects registration when name omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-010	rejects registration when email omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-011	rejects registration when password omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-012	rejects registration when name and email omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-013	rejects registration when name and password omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-014	rejects registration when email and password omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-015	rejects registration when all required fields omitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-016	rejects registration password "a" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-017	rejects registration password "12" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-018	rejects registration password "abc" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-019	rejects registration password "abcd" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-020	rejects registration password "abcde" below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-021	rejects registration password " " below minimum length	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-022	rejects unsupported registration role "superuser"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-023	rejects unsupported registration role "guest"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-024	rejects unsupported registration role "Patient!"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-025	rejects unsupported registration role "doctor-admin"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-026	rejects unsupported registration role "admin "	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-027	rejects unsupported registration role " PATIENT"	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-028	rejects unsupported registration role 7	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-029	rejects unsupported registration role {}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-030	requires both verification fields for {}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-031	requires both verification fields for {"email":"patient@example.com"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-032	requires both verification fields for {"otpCode":"123456"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-033	requires both login credentials for {}	Additional authentication input cases backend\test\auth.test.js	PASSED

Test Case ID	Test Case Name	Module / short description	Status
BE-F-034	requires both login credentials for {"email":"patient@example.com"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-035	requires both login credentials for {"password":"secret123"}	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-036	allows patient access when permitted by patient	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-037	allows doctor access when permitted by doctor	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-038	allows admin access when permitted by admin	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-039	allows patient access when permitted by doctor, patient	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-040	allows doctor access when permitted by admin, doctor	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-041	allows admin access when permitted by patient, admin	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-042	denies role access when user {"role":"patient"} is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-043	denies role access when user {"role":"doctor"} is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-044	denies role access when user {"role":"admin"} is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-045	denies role access when user undefined is not permitted	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-046	rejects protected request with no cookie or authorization header	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-047	rejects protected request with empty authentication cookie	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-048	rejects protected request with unsupported authorization scheme	Additional authentication input cases backend\test\auth.test.js	PASSED
BE-F-049	rejects protected request with malformed token	JWT authentication checks backend\test\auth.test.js	PASSED
BE-F-050	rejects protected request with token signed by an unknown secret	JWT authentication checks backend\test\auth.test.js	PASSED
BE-F-051	rejects protected request with expired token	JWT authentication checks backend\test\auth.test.js	PASSED
BE-NF-001	keeps registration validation p95 below 100ms across 50 requests	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
BE-NF-002	returns the same validation response across 50 repeated requests	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
BE-NF-003	clears the session cookie consistently across 25 logouts	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
BE-NF-004	returns consistent forbidden responses across 50 unauthorized role checks	Backend reliability/performance backend\test\nonfunctional.test.js	PASSED
AI-F-001	health endpoint returns service status	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-002	generate rejects missing prompt	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-003	generate rejects whitespace prompt	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-004	generate returns mocked local llm response	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-005	analyze report rejects missing report text	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-006	analyze report rejects whitespace report text	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-007	analyze report extracts and preserves verified metrics	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED

Test Case ID	Test Case Name	Module / short description	Status
AI-F-008	analysis rejects invalid llm schema	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-009	repair and parse json handles code fences and trailing commas	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-010	repair and parse json rejects empty llm output	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-011	analyze report falls back to deterministic result when llm is offline	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
AI-F-012	generate returns service unavailable when local llm is offline	FastAPI AI/report analysis API ai-service\tests\test_api.py	PASSED
FE-NF-SEC-01	Frontend npm production dependency audit	Production dependency security audit frontend\package-lock.json	FAILED
BE-NF-SEC-01	Backend npm production dependency audit	Production dependency security audit backend\package-lock.json	FAILED
FE-NF-QA-01	Frontend Oxlint static analysis	Static analysis / maintainability frontend\src\ (workspace)	PASSED
FE-NF-QA-02	Frontend production build	Build / release quality frontend\vite.config.js	PASSED